When Can Markets Crash Without News?

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Abstract. We characterize structural barriers, attainable losses, and execution requirements for terminal systemic drawdowns in a deterministic multiasset market with fixed fundamentals, paired-dollar trades, and zero terminal executed-dollar flow in each asset. Under the explicitly specified price-level recovery law, a response-adapted coordinate gives sufficient elasticity thresholds: $e(S) = Sg'(S)/g(S) \geq 2$ excludes positive drawdowns under our mean-log-loss and breadth criterion, while $e(S) \geq 1$ excludes a simultaneous decline of all prices. Quadratic, linear, and constant per-share responses yield distinct exclusion, inventory-bounded, and severe partial-coverage reachability regimes. The constant-response severe-loss construction relies on positive prices approaching zero and permitted instantaneous pulses; it is not an empirically calibrated crash mechanism. We derive turnover and inventory requirements, a sharp full-coverage horizon ceiling, and matching quadratic upper and lower bounds for small execution rates under finite gross execution rates. Under constant impact and bounded positive prices, an additional turnover–dissipation inequality bounds the aggregate quadratic price-potential deficit. A two-session interval-certified witness places nine of ten assets at consecutive lower closing limits and attains systemic log loss above 0.020 under specified rate and rolling-band constraints. Dimensionless resource comparisons and a nonnegative execution-cost identity clarify the economic scope of these mathematical reachability results.

Key words. market crashes; price impact; market resilience; systemic drawdown; reachability; execution constraints.

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1. Introduction.

Sharp market declines can occur without fundamental news. Order execution shifts marginal prices, and market resilience governs the decay of those displacements. We ask whether a market with fixed fundamentals can be driven into a broad systemic drawdown solely by trades that obey its settlement and recovery rules, and which trading resources are required to reach such a state.

News-free price fluctuations have been explained through several equilibrium and informational mechanisms. Dynamic hedging and inference can amplify selling pressure [15], while trading itself can reveal information without external news [24]. Nonsynchronous liquidity demand generates endogenous imbalances [16], while cross-asset liquidity feedback, distressed sales, and balance-sheet contagion can transmit losses [8, 10, 11]. Strategic predatory trading is another mechanism [6]. Transaction-level evidence on the 2010 Flash Crash documents a different, empirical object: intermediation during acute selling pressure [20]. We do not model strategic incentives, intermediary balance sheets, or an equilibrium probability of a crash; we instead characterize feasible terminal price states of a specified control system.

Quantitative price-impact models provide a framework for the execution side of this problem. Starting from Kyle's price-formation benchmark [21], resilient-liquidity models characterize execution costs and replenishment [3, 23]. General-shaped order books distinguish recovery of displaced volume from recovery of price displacement [1]; nonlinear transient propagator models study a further, distinct specification of impact and optimal execution [13]. Trading-

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cost and manipulation restrictions constrain these specifications [4, 14, 17]. Across assets, multivariate transient impact, cross-impact arbitrage restrictions, and empirical cross-impact measurement address interactions absent from our diagonal price-response model [2, 5, 25]; strategic multiasset impact games provide another comparison [12]. Metaorder evidence for a linear-to-square-root impact crossover concerns cumulative impact versus order size, not directly the local state elasticity in our model [7]. Kyle and Obizhaeva [22] relate large bets and execution rates to crash-scale observations. Here we hold the execution law fixed and ask a different question: which terminal loss states does it permit?

We consider a cashless reallocation system of risky assets in which every executed sale is paired with an equal-dollar purchase, and each asset has zero net executed dollar flow at the horizon. Prices recover toward a fixed benchmark F, while executed shares shift marginal prices through a positive response function $g(S)$. Our central analytical tool is the response-adapted generalized price coordinate

$$\Phi(S) = \int_F^S \frac{u}{g(u)}\,\mathrm{d}u. \tag{1.1}$$

In this coordinate, each executed dollar produces one unit of coordinate displacement. Paired trading therefore conserves the sum of these coordinates, while price resilience drives that sum between trades. The curvature of Φ yields global exclusion barriers and separates three response regimes under common constraints.

Beyond the elasticity barriers, the main analytical results for constant impact are severe partial-coverage reachability under uniform finite resources, the sharp full-coverage horizon ceiling, and the quadratic small-rate law. Section 5 additionally relates terminal quadratic price-potential dissipation to gross executed-dollar turnover in a bounded price region. The multiday analysis then examines simultaneous feasibility under execution constraints: receiving assets must absorb paired-dollar inflows within their own rate caps and price bands, while their appreciation reduces mean log loss.

Overview of main results. Under the paired-dollar, resilient market defined in Section 2, the following conclusions hold.

(i) *Structural exclusion.* If $e(S) = Sg'(S)/g(S) \geq 2$ at all admissible prices, then $\mathcal{C}_\alpha(t) \leq 0$ for every breadth and time. The weaker condition $e(S) \geq 1$ excludes a simultaneous decline of all assets.

(ii) *Reachability regimes.* With $g_p(S) = \eta(S/F)^p$, $p = 2, 1, 0$ respectively produce exclusion, inventory-bounded partial-coverage drawdowns, and, for $k = \lceil \alpha N \rceil < N$ and sufficiently long finite horizons, arbitrarily severe partial-coverage losses under fixed finite turnover and inventory caps when pulses are allowed. For constant impact and $k = N$, a sharp horizon-dependent bound replaces unrestricted severity.

(iii) *Execution resources.* General positive g yields explicit necessary dollar-turnover and share-inventory bounds. With constant per-share impact and bounded gross executed-share rates, the small-rate crisis bound is quadratic and this order is attained by a continuous dollar-closed construction.

(iv) *Multiday execution.* Rolling price bands restrict per-session losses, while rate limits and the receiving assets' capacity restrict consecutive lower-limit closes. A computer-assisted

two-day ten-asset path satisfies these controls, has nine repeated lower-limit closes, and yields $\mathcal{C}_{0.9}(T) > 0.020$; for that instance, $0.044 < \bar{v}_* < 0.285$.

2. The market: assumptions, execution, and price dynamics. We specify a risky-asset reallocation submarket in which every executed sale is matched by an equal-dollar purchase. Settlement and price response are distinct assumptions: the former constrains flows, while the latter determines the induced price motion. All subsequent theorems use the same market definition.

Assumption 2.1 (The resilient paired-dollar market). Fix $N \geq 2$ assets and a finite horizon $[0, T]$. The following jointly define the admissible market.

(M1) **Positive prices and coordinates.** Each transaction price $S_i(t)$ is strictly positive and initially $S_i(0) = F > 0$. A strictly increasing C^1 coordinate $q_i = \phi_i(S_i)$ preserves price priority. The analysis concerns deterministic admissible execution paths.

(M2) **Trade and execution accounting.** Executed share change is the signed quantity $\mathrm{d}\theta_i$; unsigned volume counts its absolute increments, not both counterparties twice. Executions obey marginal-price settlement $\mathrm{d}X_i = S_i\,\mathrm{d}\theta_i$. Matching uses price priority conditional on an order book; the effective price response is a separately imposed, positive, locally C^2 function $g(S)$, such that $\mathrm{d}S_i|_{\mathrm{trade}} = g(S_i)\,\mathrm{d}\theta_i$.

(M3) **No fundamental news and resilience.** The benchmark F stays fixed. Between trades, prices recover toward F at a common constant rate $\rho > 0$: $\dot{S}_i = -\rho(S_i - F)$ in the absence of execution. The response function g and ρ are specified exogenously.

(M4) **Paired-dollar transaction network.** Each permitted transaction moves an equal executed-dollar amount between a source asset and a destination asset: $\mathrm{d}\boldsymbol{X} = B\,\mathrm{d}\boldsymbol{z}$, where B is the oriented incidence matrix of the complete bidirectional graph, $\mathrm{d}\boldsymbol{z} \geq 0$, and $\mathbf{1}^\top B = 0$. This condition specifies an idealized paired-reallocation submarket without external cash inflows.

(M5) **Admissible controls and terminal closure.** Paths are piecewise continuous and of finite execution variation, with finitely many marginally settled instantaneous pulses allowed; every price remains positive and required sale inventory is available without short sales. The terminal comparison imposes $X_i(T) = 0$ for *each* asset, not $\theta_i(T) = \theta_i(0)$. Where resource caps are claimed, identical total paired-dollar turnover and per-asset starting share limits are imposed across markets.

Price-level resilience as a modeling choice. Assumption (M3) imposes $\dot{S} = -\rho(S - F)$ *in price levels*. It is not derived from the frozen-book matching relation in (M2). For comparison, a displacement-in-volume coordinate $V(S) = \int_F^S g(u)^{-1}\,\mathrm{d}u$, relaxing according to $\dot{V} = -\rho V$, instead implies

$$\dot{S}\big|_{\mathrm{vol}} = -\rho g(S)V(S), \qquad \dot{\Phi}(S)\big|_{\mathrm{vol}} = -\rho S V(S).$$

Except in special response families such as constant g, this is a different restoring field from $-\rho S(S - F)/g(S)$ in (2.6). The two forms are conceptually related to the distinct resilience specifications discussed by Alfonsi, Fruth, and Schied [1]; they need not share the same invariant-region barriers. The elasticity thresholds proved below are statements about the *joint* price-response and price-level recovery assumptions of this paper, not universal thresholds for every resilient order-book model.

Interpretation of terminal dollar closure. The complete two-sided graph in (M4) represents a cashless portfolio-switching pool in which executed sale proceeds immediately purchase other risky assets. In ordinary exchange-wide order flow, a cash account or additional counterparties may absorb those proceeds. The terminal condition $X_i(T) = \int_0^T S_i \, \mathrm{d}\theta_i = 0$ isolates residual price displacement after each asset's net executed dollar flow has canceled. A standard share-flat round trip instead requires $\theta_i(T) = \theta_i(0)$. By (2.11), dollar closure with terminal displacement or positive recovery dissipation consumes initially available share inventory: $F[\theta_i(0) - \theta_i(T)] = E(S_i(T)) + \rho \int_0^T (S_i - F)^2 / g(S_i) \, \mathrm{d}t$. Thus dollar closure can leave different terminal shareholdings and does not imply zero execution cost. The nonnegative value of the consumed inventory at the unchanged fundamental benchmark is interpreted in Section 8; assigning profits to a particular counterparty would require further balance-sheet assumptions.

Finite-variation paths and marginally settled pulses. For absolutely continuous controls the differential notation is classical. At a jump we do *not* use a left-endpoint or right-endpoint Stieltjes product $S_i \Delta\theta_i$. Instead physical time is held fixed and each directed edge $e : i \to j$ with dollar mass $z > 0$ is executed along a specified within-event parameter $a \in [0, z]$:

$$\frac{\mathrm{d}\widetilde{S}_i}{\mathrm{d}a} = -\frac{g(\widetilde{S}_i)}{\widetilde{S}_i}, \quad \frac{\mathrm{d}\widetilde{S}_j}{\mathrm{d}a} = \frac{g(\widetilde{S}_j)}{\widetilde{S}_j}, \quad \frac{\mathrm{d}\widetilde{\theta}_i}{\mathrm{d}a} = -\frac{1}{\widetilde{S}_i}, \quad \frac{\mathrm{d}\widetilde{\theta}_j}{\mathrm{d}a} = \frac{1}{\widetilde{S}_j}. \tag{2.1}$$

The resulting endpoint flows satisfy $\Delta X_i = -z$, $\Delta X_j = z$ and $\Delta\Phi(S_i) = -z$, $\Delta\Phi(S_j) = z$. Equivalently, a single-share impulse is evaluated by solving $\mathrm{d}\widetilde{S}/\,\mathrm{d}\xi = g(\widetilde{S})$ and integrating $\int \widetilde{S} \, \mathrm{d}\xi$ along that response trajectory. Jumps are thus *flow-completed* (Marcus-type) updates preserving the ordinary chain rule. Finite pulses on distinct edges are processed in their prescribed order. A pulse is admissible only while all intermediate prices remain strictly positive. These rules specify the interpretation of all jump expressions in what follows.

From the conditional matching map to an effective response. Fix a frozen book $\mathcal{B}$ with locally positive continuous share-depth density $\rho_S(S; \mathcal{B})$. A one-sided sweep whose signed marginal share size is $\mathrm{d}\theta$ obeys the local relation

$$\mathrm{d}\theta = \rho_S(S; \mathcal{B}) \, \mathrm{d}S, \qquad \rho_q(q; \mathcal{B}) \, \mathrm{d}q = \rho_S(S; \mathcal{B}) \, \mathrm{d}S. \tag{2.2}$$

The exact discrete version uses the generalized inverse of the book's cumulative depth. In the continuous approximation, changing $q = \phi(S)$ changes the density numerically through the Jacobian, not the matching priority. The relation refers to the marginal executed price. A dynamic midpoint, replenished book, and interasset cross-impact are *not* derived by coordinate transformation. Assumption (M2) instead chooses $g(S) = 1/\rho_S^{\mathrm{eff}}(S)$ as an *effective*, state-dependent response for the rest of the paper.

Price dynamics for N assets. Under the full market specification, the level-price law is

$$\boxed{\mathrm{d}S_i(t) = -\rho\big(S_i(t) - F\big) \, \mathrm{d}t + g(S_i(t)) \, \mathrm{d}\theta_i(t), \qquad \mathrm{d}X_i = S_i \, \mathrm{d}\theta_i.} \tag{2.3}$$

Choose the strictly monotone generalized coordinate $q_i = \log(S_i/F)$. Conversion to paired dollars gives

$$\mathrm{d}q_i\big|_{\mathrm{trade}} = \frac{g(S_i)}{S_i^2} \, \mathrm{d}X_i. \tag{2.4}$$

Combining this conversion, transaction pairing, and resilience yields the full multiasset system

$$\boxed{\mathrm{d}\boldsymbol{q} = -\rho(\mathbf{1} - e^{-\boldsymbol{q}})\,\mathrm{d}t + D(\boldsymbol{q})B\,\mathrm{d}\boldsymbol{z}, \qquad D_{ii}(\boldsymbol{q}) = \frac{g(Fe^{q_i})}{F^2 e^{2q_i}}.} \tag{2.5}$$

The exponential acts componentwise. This log coordinate is convenient for measuring losses, but executed dollars admit a simpler *response-adapted* coordinate. Let $y_i = \Phi(S_i)$ with Φ as in (1.1). It is strictly increasing and satisfies $\Phi(F) = 0$. Moreover, among differentiable coordinates ψ normalized by $\psi(F) = 0$, the requirement that each marginal trade obey $\mathrm{d}\psi(S)|_{\mathrm{trade}} = \mathrm{d}X$ forces $\psi'(S)g(S) = S$ and thus uniquely selects $\psi = \Phi$. Define the restoring response

$$\mathcal{R}(y) = \frac{S(S-F)}{g(S)}, \qquad S = \Phi^{-1}(y).$$

The entire multiasset market then has the *trade-normalized form*

$$\boxed{\mathrm{d}\boldsymbol{y} = B\,\mathrm{d}\boldsymbol{z} - \rho\mathcal{R}(\boldsymbol{y})\,\mathrm{d}t, \qquad \mathbf{1}^{\top}B = 0,} \tag{2.6}$$

where $\mathcal{R}$ acts componentwise and the execution rule applies also to flow-completed pulses. Trading is therefore an additive zero-sum input; all nonlinear price-state dependence has moved into the restoring term and the inverse map $S = \Phi^{-1}(y)$. The transformed state space is $\mathcal{J}^N$, where $\mathcal{J} = \Phi((0,\infty))$. For $g_p(S) = \eta(S/F)^p$, the range is $(-F^2/[\eta(2-p)], \infty)$ for $p < 2$, $\mathbb{R}$ for $p = 2$, and $(-\infty, F^2/[\eta(p-2)])$ for $p > 2$. Thus additive trading inputs do not remove strict price positivity or execution constraints. This change of coordinates does not alter which price states are reachable. With $\rho = 0$ the traded sum of generalized coordinates is exactly conserved.

Definition 2.2 (Terminal systemic drawdown). *Let $L_i(T) = \log[F/S_i(T)]$ and $\overline{L}(T) = N^{-1}\sum_i L_i(T)$. The kth-largest loss, where $k = \lceil \alpha N \rceil$, is $Q_\alpha(T)$. Define*

$$\boxed{\mathcal{C}_\alpha(T) = \min\{\overline{L}(T), Q_\alpha(T)\}.} \tag{2.7}$$

For $\delta > 0$, a terminal crisis *occurs when $\mathcal{C}_\alpha(T) \geq \delta$. Thus the mean loss and at least k individual losses must each reach δ. The target is selected independently of the price-response function.*

The mean-loss component measures the decline of the equally weighted geometric price index, since

$$\overline{L}(T) = -\log\left[\left(\prod_{i=1}^{N}\frac{S_i(T)}{F}\right)^{1/N}\right].$$

The criterion therefore combines geometric-index deterioration with a breadth requirement. Arithmetic price averages and market-capitalization-weighted indices are different aggregates; for example, the $p = 1$ market below can have $\mathcal{C}_\alpha(T) > 0$ while $\sum_i S_i(T) = NF$.

The discrete coverage rule is material: for $N = 3$, $\alpha = 0.9$ means $k = 3$, so it requires a fall in all three assets. More generally, the contrast $k = N$ versus $k < N$ determines

whether a compensating asset exists at all. This same distinction appears first in the A/B/C classification and later in the multiday cross-asset capacity inequality. A first-passage event $\tau_{\alpha,\delta} = \inf\{t : \mathcal{C}_\alpha(t) \geq \delta\}$ is different from a *terminal* crisis under dollar closure; we study the latter.

Definition 2.3 (Turnover, share inventory and terminal critical budget). *For an admissible path γ, set*

$$M(\gamma) = \sum_e \int_0^T \mathrm{d}z_e, \qquad H_i^-(\gamma) = \int_0^T \big(-\mathrm{d}\theta_i\big)_+. \tag{2.8}$$

Under allowed initial inventory H_i and terminal dollar closure, define

$$M_*^{\mathrm{end}} = \inf\{M(\gamma) : X_i(T) = 0\ \forall i,\ \mathcal{C}_\alpha(T) \geq \delta,\ \gamma\ \textit{admissible}\}, \tag{2.9}$$

with $\inf \varnothing = +\infty$. A paired dollar transfer is counted once in M, although it appears with opposite signs in two asset ledgers.

The precise no-profitable-round-trip property. Define

$$E(S) = \int_F^S \frac{u - F}{g(u)}\,\mathrm{d}u \geq 0. \tag{2.10}$$

For any admissible path beginning at F the execution-cost identity is

$$\boxed{\int_0^T S_i\,\mathrm{d}\theta_i = F[\theta_i(T) - \theta_i(0)] + E(S_i(T)) + \rho\int_0^T \frac{(S_i - F)^2}{g(S_i)}\,\mathrm{d}t.} \tag{2.11}$$

Thus a *share-flat* round trip has nonnegative executed cost under (M2)–(M3). The paper's terminal condition $X_i(T) = 0$ instead closes executed *dollars*, not shares: a terminally dollar-flat path may consume initial inventory. Appendix A proves the accounting identity and its consequences for admissible controls.

Comparison with share-flat trading. The terminal dollar condition and the standard share-flat condition are not interchangeable. For example, with $g \equiv \eta$, a single asset purchased by a sufficiently small $a > 0$ shares at time 0 and sold by a shares at time T has $\theta(T) = \theta(0)$ but terminal price $S(T) = F - \eta a(1 - e^{-\rho T}) < F$; its net executed dollars are not zero and the positive cost in (2.11) is paid through the cash account. Conversely, under (M4), requiring every asset to be share-flat *as well as* terminally dollar-flat would force $\sum_i [E(S_i(T)) + \rho\int_0^T (S_i - F)^2/g(S_i)\ \mathrm{d}t] = 0$. Thus there can be no positive-time dissipative price excursion under both closures together. The dollar-flat protocol deliberately isolates an economic question distinct from standard share-round-trip execution.

3. Structural barriers to market crashes.

The normal form (2.6) supplies a conserved scalar during trading. Write $H(\boldsymbol{S}) = \mathbf{1}^\top \boldsymbol{y} = \sum_i \Phi(S_i)$. Summing over assets cancels every matched dollar transfer and leaves only recovery:

$$\boxed{\mathrm{d}H = -\rho\sum_{i=1}^N \mathcal{R}(y_i)\,\mathrm{d}t = -\rho\sum_{i=1}^N \frac{S_i(S_i - F)}{g(S_i)}\,\mathrm{d}t.} \tag{3.1}$$

The scalar H is invariant under trading and evolves through price recovery between trades. Whether the recovery drift can carry H through a crisis boundary depends on the curvature of Φ. Introduce the response elasticity

$$e(S) = \frac{Sg'(S)}{g(S)}. \tag{3.2}$$

The derivatives of Φ and $G(x) = \Phi(Fe^x)$ are

$$\Phi''(S) = \frac{1 - e(S)}{g(S)}, \qquad G''(x) = \frac{S^2}{g(S)}\,[2 - e(S)], \quad S = Fe^x. \tag{3.3}$$

The distinction between these two curvatures tracks the definition of systemic loss. A barrier to *every asset* being below its benchmark is weaker than a barrier to our functional, which also involves the *average* log loss. Thus the same coordinate can produce two different exclusion statements.

Theorem 3.1 (Barrier to a decline of every asset). *If $e(S) \geq 1$ at all prices attainable by admissible paths, then $S_i(t) < F$ for every i is impossible at any time t. In particular, no positive crisis with $k = N$ is reachable.*

Proof. Equation (3.3) makes Φ concave. Its tangent inequality at S is $\Phi(F) \leq \Phi(S) + \Phi'(S)(F - S)$, hence $(S - F)\Phi'(S) \leq \Phi(S)$. Consequently (3.1) implies $\dot{H} \geq -\rho H$ between trades; pulses preserve H. Since $H(0) = 0$, comparison gives $H(t) \geq 0$. If every $S_i(t) < F$, strict monotonicity and $\Phi(F) = 0$ instead imply $H(t) < 0$, a contradiction. ∎

Theorem 3.2 (Barrier to a systemic crisis of any breadth). *If $e(S) \geq 2$ on the admissible price range, then for every admissible control, $N \geq 2$, $\alpha \in (0, 1]$ and $t \in [0, T]$,*

$$\boxed{\mathcal{C}_\alpha(t) \leq 0.} \tag{3.4}$$

Thus no strictly positive systemic terminal-crisis threshold can be reached.

Proof. The preceding theorem gives $H(t) \geq 0$. The second identity in (3.3) makes $G(x) = \Phi(Fe^x)$ concave and strictly increasing. Jensen's inequality yields

$$0 \leq \frac{H(t)}{N} = \frac{1}{N}\sum_i G(q_i(t)) \leq G\left(\frac{1}{N}\sum_i q_i(t)\right).$$

As $G(0) = 0$ and G is strictly increasing, $\sum_i q_i(t) \geq 0$. Hence the mean loss $\overline{L}(t) = -N^{-1}\sum_i q_i(t)$ is nonpositive; Definition 2.2 implies (3.4). ∎

The two theorems locate states that *no* admissible trade program can reach under their respective assumptions. They provide global *sufficient* conditions for excluding the specified loss states. Neither exclusion requires terminal dollar closure. If the inequalities hold only near F, the exclusions apply only while the entire path stays there; limited inventory, trading links, or time may create further barriers. The A/B/C comparison establishes constructive reachability results within one response family when these structural conditions are relaxed. Crucially, the tangent comparison in Theorem 3.1 invokes the drift $-\rho S(S - F)/g(S)$; substituting a different volume-recovery drift is not covered by either theorem. We do not claim the corresponding thresholds are invariant to that modeling change.

4. Three market regimes under common constraints. To compare exclusion and reachability under matched assumptions, we vary only the response shape in the family

$$g_p(S) = \eta \left(\frac{S}{F}\right)^p, \qquad p \in \{2, 1, 0\}, \quad \eta > 0. \tag{4.1}$$

All three markets have identical N, T, F, ρ, η, identical initial prices and initial *impact*, the same paired-dollar trading network, and, when compared numerically, the same turnover and inventory caps. They differ only in the exponent p. In the log-dollar equation (2.5),

$$D_{ii}^{(p)}(q_i) = \frac{\eta}{F^2} e^{(p-2)q_i}, \qquad e(S) = p. \tag{4.2}$$

The same coordinate formula integrates explicitly for every exponent:

$$\Phi_p(S) = \begin{cases} \dfrac{F^p}{\eta(2-p)}\left(S^{2-p} - F^{2-p}\right), & p \neq 2, \\ \dfrac{F^2}{\eta} \log(S/F), & p = 2. \end{cases}$$

Consequently, paired trades conserve a quantity proportional to $\sum_i S_i^2$, $\sum_i S_i$, or $\sum_i \log S_i$ for $p = 0, 1, 2$, respectively. Recovery evolves these quantities between trades. Figure 1 shows the corresponding instantaneous response per executed dollar: below F, the $p = 0$ response amplifies dollar trades most strongly.

Theorem 4.1 (A/B/C reachability regimes). *Let $N \geq 2$ and $k = \lceil \alpha N \rceil < N$. Compare the three markets in* (4.1) *under the same model protocol.*

(A) ***Barrier market*** *($p = 2$)**:*** *$\mathcal{C}_\alpha(t) \leq 0$ for every admissible path and every horizon.*

(B) ***Inventory-bounded market*** *($p = 1$)**:*** *positive but sufficiently small terminal systemic drawdowns are reachable. If every asset starts with a fixed finite inventory cap H_0 and short selling is forbidden, the terminal crisis severity is bounded above independently of the trade sequence.*

(C) ***Market with unbounded partial-coverage losses*** *($p = 0$)**:*** *for every fixed $N \geq 2$ and $k < N$, there exists a sufficiently long but finite horizon T and finite constants M_0, H_0 such that, for* every *finite $\delta > 0$, an admissible positive-price path with $M \leq M_0$, $H_i^- \leq H_0$, and $X_i(T) = 0$ attains $\mathcal{C}_\alpha(T) > \delta$.*

The constants in (C) may depend on N, T, F, ρ, η, but not on δ. They do not remain uniform as $N \to \infty$.

Proof. (A) follows from Theorem 3.2. (B) follows from the two-pulse construction in Proposition 4.3 and the inventory inequality (4.3) below. (C) is established constructively in Appendix C: $N - 1$ targets are driven arbitrarily close to zero, while one asset serves as a cash-flow compensator. The control uses positive prices, fully paired marginal-dollar executions, a partition of recovery time, and terminal closure of every X_i. Its explicit turnover and inventory bounds are independent of the target final price $\varepsilon \downarrow 0$. ∎

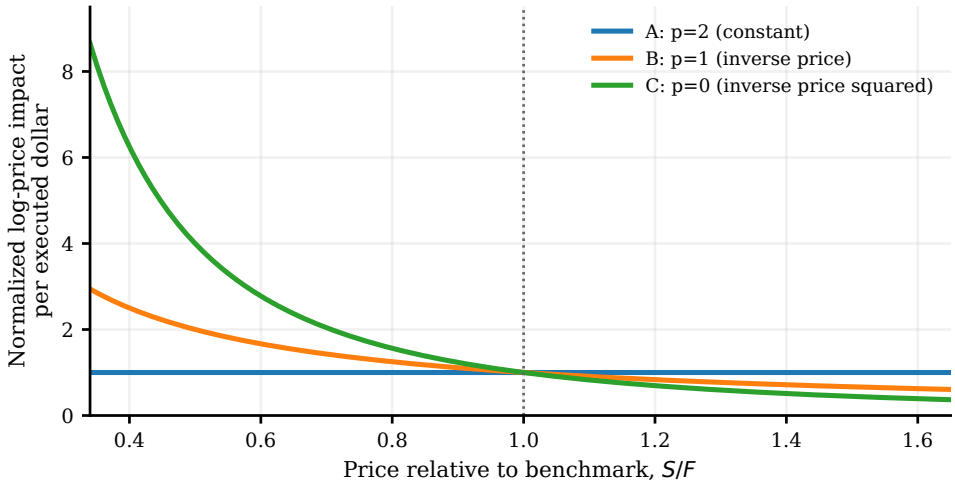


Figure 1. ***A single response family, three price sensitivities.*** *The plot shows the model-implied instantaneous* log-price change per executed dollar*, normalized by its common initial value. All curves coincide at* $S = F$*.*

Under the same executed-dollar and recovery protocol, the three response functions produce distinct attainable-state geometries. Market A supplies a global invariant-region barrier; Market B shows that nonzero partial-coverage losses need not permit arbitrarily large ones when stock inventory is finite; and Market C supplies an explicit trajectory with a severe loss. The condition $k < N$ in (C) is essential: at least one asset must be available to carry the compensating flow. Finite gross share rates further constrain the compensating asset's *capacity*.

For $p = 1$, the inventory potential from (2.10) is

$$E_1(Fe^{-L}) = \frac{F^2}{\eta}\left(L + e^{-L} - 1\right). \tag{4.3}$$

By (2.11), terminal dollar closure and no short selling imply $E_1(S_i(T)) \le FH_0$. Since $L + e^{-L} - 1 \to \infty$ as $L \to \infty$, no fixed finite inventory can support arbitrarily severe losses of the requisite k assets. By contrast, for $p = 0$, $E_0(S) = (S - F)^2/(2\eta)$ remains bounded as $S \downarrow 0$; this makes the construction in (C) possible but does not by itself prove reachability.

Proposition 4.2 (Local all-asset decline). *If g is C^3 near F and $e(F) < 1$, then for every $N \ge 2$ there exists a terminally dollar-closed, positive-price, small-amplitude three-pulse control under which* every *$S_i(T) < F$. With $c = g(F)/F^2$ and $r = e^{-\rho T/2}$, the losses of a zero-sum impulse direction $\boldsymbol{u}$ with every $u_i \neq 0$ satisfy*

$$L_i(T) = \frac{1 - e(F)}{2} c^2 h^2 u_i^2 r(1 - r^2) + O(h^3) > 0 \tag{4.4}$$

for sufficiently small $h > 0$.

Proof. Apply the executed-dollar pulses $h\boldsymbol{u}$, $-(1 + r)h\boldsymbol{u}$, $rh\boldsymbol{u}$ at $0, T/2, T$. They are pairwise dollar-balanced and sum to zero for every asset. Taylor expansion in the coordinate $y = \Phi(S)$ gives (4.4); details appear in Appendix B. ∎

Proposition 4.3 (Local partial-coverage decline). *Let $k < N$, $e_0 = e(F) < 2$, $c = g(F)/F^2$, and $r = e^{-\rho T} < 2 - e_0$. Select a zero-sum dollar direction with at least k positive components and execute $h\boldsymbol{u}$ at time 0 and $-h\boldsymbol{u}$ at time T. Then $\mathcal{C}_\alpha(T) > 0$ for all sufficiently small $h > 0$, and*

$$\overline{L}(T) = \frac{c^2 h^2}{2N}(1-r)(2-e_0-r)\sum_i u_i^2 + O(h^3). \tag{4.5}$$

Proof. The individual losses satisfy $L_i(T) = c(1-r)hu_i + O(h^2)$, while the first-order term vanishes when averaged because $\sum_i u_i = 0$. The strictly positive quadratic coefficient in (4.5) proves the result. See Appendix B. ∎

The preceding two propositions describe local reachability within this classification. A two-pulse path can generate a partial-coverage loss under a less restrictive elasticity threshold; a three-pulse path with an additional cancellation step can generate a small decline of *every* asset when $e(F) < 1$. The global severe-loss construction for Market C is established separately.

A concrete common-budget comparison. For $N = 3$, $k = 2$, $F = \eta = \rho = 1$, $T = 3$, common turnover cap $M_0 = 3$, and per-asset initial inventory cap $H_0 = 8.2$, the $p = 2$ market cannot reach any positive crisis; the $p = 1$ market can reach a small positive crisis but not the threshold $\delta = 10$ (it would require over 9 shares of selling by any qualifying asset); a constructive $p = 0$ path reaches $\mathcal{C}_{2/3}(T) \approx 13.32$ with turnover approximately 2.62 and a maximum per-asset cumulative share sale below 3.94. Here the large threshold $\delta = 10$ serves as a theoretical stress test to separate bounded from unbounded logarithmic losses, not as an empirically representative market crash. In Market C, $E_0(Fe^{-\delta}) = \frac{F^2}{2\eta}(1-e^{-\delta})^2$ stays finite as $\delta \to \infty$, whereas $\log(F/S) \to \infty$ as $S \downarrow 0$. This difference is central to the extreme-severity conclusion: all constructed paths retain strictly positive prices, but the limiting argument approaches a boundary at which a constant additive per-share response should not be read as a calibrated market-depth model. The choices $F = \eta = 1$ normalize units and do not assert that the observed market permits these flows relative to daily volume. Validation code accompanies the source files.

The common trading experiment across the three response regimes is illustrated in Figure 2.

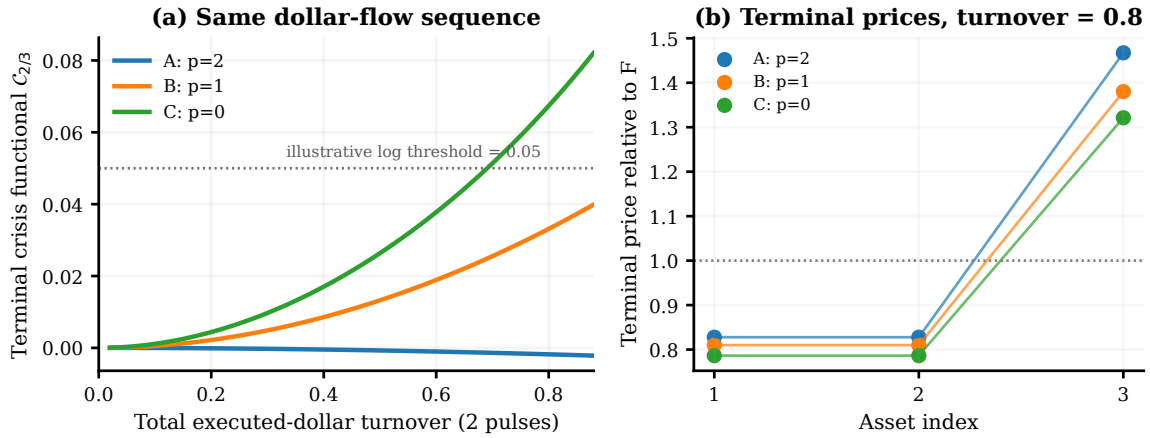


Figure 2. ***An identical trading experiment across A/B/C response functions.*** *Two terminally offsetting executed-dollar pulses act on $N = 3$ assets with $k = 2$, $F = \eta = \rho = 1$, and $T = 3$. Left: the measured* terminal *crisis functional as paired-dollar turnover increases; right: terminal prices under the same total turnover $M = 0.8$. The dotted horizontal line marks the reference log-loss threshold 0.05.*

The continuous power-law spectrum. The three canonical markets in Theorem 4.1 are part of a continuous family: retain the expression for g_p in (4.1), now allowing $0 \le p \le 2$. Since $e(S) = p$, the elasticity barriers and the local constructions distinguish two critical exponents. In geometric terms, $\Phi_p''(S) = (1-p)/g_p(S)$, while $G_p''(q) = F^2(2-p)e^{(2-p)q}/\eta$ for $G_p(q) = \Phi_p(Fe^q)$: the two curvatures reach their respective thresholds at $p = 1$ and $p = 2$. For $0 \le p < 2$, the dollar-coordinate distance to zero is finite, whereas the share-coordinate distance changes character at $p = 1$:

$$-\Phi_p(0^+) = \frac{F^2}{\eta(2-p)}, \qquad -\Psi_p(0^+) = \begin{cases} \dfrac{F}{\eta(1-p)}, & 0 \le p < 1, \\ +\infty, & 1 \le p < 2. \end{cases}$$

The inventory potential in (2.10), evaluated at $S = Fe^{-L}$, is for $p \neq 1, 2$

$$E_p(Fe^{-L}) = \frac{F^2}{\eta}\left[\frac{e^{(p-1)L}-1}{p-1} - \frac{1-e^{-(2-p)L}}{2-p}\right].$$

For $0 \le p < 1$, its zero-price limit is $E_p(0^+) = F^2/[\eta(1-p)(2-p)] < \infty$. Proposition 4.2 also makes a small, simultaneous decline of every asset locally reachable in this regime. Finiteness of the coordinate distances and inventory potential, however, does not by itself extend the *uniform-budget, arbitrarily severe* construction in Theorem 4.1(C) beyond the proven $p = 0$ case.

At $p = 1$, Theorem 3.1 excludes simultaneous declines, while Proposition 4.3 gives local partial-coverage losses at every positive horizon. The potential (4.3) then grows linearly with L. For $1 < p < 2$, simultaneous declines remain excluded, but local partial-coverage losses are reachable under the sufficient horizon condition $e^{-\rho T} < 2 - p$, or $T > -\log(2-p)/\rho$. Here

the dollar distance is still finite while $E_p(Fe^{-L}) \sim F^2 e^{(p-1)L}/[\eta(p-1)]$ as $L \to \infty$. Indeed, if terminal executed-dollar flows vanish, short sales are prohibited, and the initial inventory of a qualifying asset is at most H_0, (5.2) implies

$$L \le \frac{1}{p-1} \log\left[1 + (p-1)\left(\frac{\eta H_0}{F} + \frac{1}{2-p}\right)\right], \qquad 1 < p < 2.$$

Thus, for fixed $p \in (1,2)$, the inventory-based severity ceiling scales at most logarithmically in large H_0, rather than linearly as for $p = 1$. At the second threshold $p = 2$, $-\Phi_2(0^+)$ diverges and Theorem 3.2 excludes every positive systemic crisis measure. These thresholds distinguish local reachability, boundary resource behavior, and the separate global severe-loss theorem; they are not claims of uniform attainability under fixed execution constraints.

4.1. A sharp boundary when every asset must fall. Market C relies on $k < N$. To see why the breadth condition matters beyond a technical assumption, keep *the same constant-impact response* but change the target to $k = N$. There is then no nonqualifying asset available to compensate the other $N - 1$ terminal losses. Rather than an arbitrary-severity construction, the model now yields an exact horizon-dependent ceiling, even before a fixed execution budget is imposed.

Theorem 4.4 (Sharp all-asset horizon threshold). *Under Assumption 2.1 with $g(S) = \eta$ and $k = N$, define*

$$\boxed{\delta_{\mathrm{crit}}(N,T) = \log \frac{\sqrt{N}}{1 + (\sqrt{N}-1)e^{-\rho T}}.} \tag{4.6}$$

For every $\delta \in (0, \delta_{\mathrm{crit}}(N,T))$, some admissible terminally dollar-closed control reaches $\mathcal{C}_1(T) \ge \delta$ with finite turnover and sufficiently large finite inventory. If $\delta \ge \delta_{\mathrm{crit}}(N,T)$, no such positive-price path exists. Thus δ_{crit} is the unattained supremum *of the all-asset common log drawdown.*

Proof sketch. Set $R(t)^2 = \sum_i S_i(t)^2$. Under constant response, any paired-dollar trade satisfies $\mathrm{d}R^2|_{\mathrm{trade}} = 2\eta \sum_i \mathrm{d}X_i = 0$. Between trades,

$$\dot{R} = -\rho\left(R - F\frac{\sum_i S_i}{R}\right).$$

For positive prices, $\sum_i S_i/R > 1$, whence $R(T) > F\{1 + (\sqrt{N}-1)e^{-\rho T}\}$. If each asset ends below $Fe^{-\delta}$, then $R(T) \le F\sqrt{N}e^{-\delta}$, proving necessity. For sufficiency, rotate the price vector through N positive spherical directions with one high coordinate, allotting each asset equal recovery weight. Each reorientation is a zero-sum paired-dollar pulse, and paired controls during recovery keep the direction fixed. Equal terminal prices and equal recovery integrals, together with $\sum_i X_i(T) = 0$, give $X_i(T) = 0$ for every i. Moving the positive low coordinates toward zero approaches the bound continuously. A full constructive proof is in Appendix D. ■

5. Necessary execution resources. The response regimes determine whether a loss state is reachable. We next quantify the dollar turnover and share sales required to attain a specified breadth and severity. The following bounds apply to general positive response functions.

Theorem 5.1 (Execution-resource lower bounds). *Suppose an admissible terminal path satisfies $\mathcal{C}_\alpha(T) \geq \delta > 0$. Write $k = \lceil \alpha N \rceil$ and $s = Fe^{-\delta}$. Alongside Φ, define the monotone share-response coordinate $\Psi(S) = \int_F^S \mathrm{d}u/g(u)$. Then the necessary dollar-turnover and total share-sale bounds are simply the respective coordinate distances to s:*

$$\boxed{M \geq -k\Phi(s) = k\int_s^F \frac{u}{g(u)}\,\mathrm{d}u, \qquad \sum_i H_i^- \geq -k\Psi(s) = k\int_s^F \frac{\mathrm{d}u}{g(u)}.} \tag{5.1}$$

Moreover, if $X_i(T) = 0$, no short sales are allowed, and the initial inventory of any qualifying asset does not exceed H_i, then

$$\boxed{\int_s^F \frac{F-u}{g(u)}\,\mathrm{d}u \leq FH_i.} \tag{5.2}$$

These necessary lower bounds need not be sharp.

Proof. For a terminally qualifying asset, the final price is at most $s < F$. Both transformed coordinates increase with price, and recovery acts toward their zero values at F. Consequently the cumulative negative executed-dollar and share variations must be at least $-\Phi(s)$ and $-\Psi(s)$, respectively. Summing over k distinct qualifying assets proves (5.1) because sales by qualifying assets form a subset of total sales, and each paired execution is counted once in total sales. Equation (5.2) follows from the nonnegative right-hand side of (2.11) with $X_i(T) = 0$ and $\theta_i(0) - \theta_i(T) \leq H_i$. The detailed variation argument is given in Appendix A. ∎

Geometrically, the resource bounds measure distances in the two monotone coordinates. Trading direction, recovery time, and the buying capacity of receiving assets impose further requirements; we turn to these execution restrictions next.

Replicating identical asset depths at fixed α and δ gives $k = \Theta(N)$, and hence $M_*^{\mathrm{end}} = \Omega(N)$ for reachable targets. This experiment holds depth per asset fixed, so aggregate depth grows with N.

The low-price integrals discriminate different kinds of resource scarcity:

$$\int_0^F \frac{u}{g(u)}\,\mathrm{d}u, \qquad \int_0^F \frac{\mathrm{d}u}{g(u)}. \tag{5.3}$$

For a local power law $g(u) \asymp u^p$, the dollar integral converges only if $p < 2$, and the share integral only if $p < 1$. These integral criteria separate resource feasibility from the structural and dynamic conditions for reachability.

Dimensionless execution scales. Let $g_F = g(F) > 0$ and define the *local benchmark dollar-impact scale* $D_F = F^2/g_F$. At the benchmark, a signed executed-dollar amount D_F corresponds to a unit first-order log-price displacement under the model's marginal response rule; this is a local scale, not a measured market depth or average daily turnover. The dimensionless controls are

$$\widehat{M} = \frac{M}{D_F}, \qquad \widehat{H}_i = \frac{g_F H_i}{F}, \qquad \widehat{v} = \frac{g_F \bar{v}}{\rho F}, \qquad \widehat{T} = \rho T.$$

For the power family $g_p(S) = \eta(S/F)^p$, the necessary dollar-turnover bound in (5.1) becomes

$$\widehat{M} \geq k \begin{cases} \dfrac{1 - e^{-(2-p)\delta}}{2-p}, & p \neq 2, \\ \delta, & p = 2. \end{cases}$$

In the ten-asset, two-session constant-impact certificate, $F = g_F = 1$, $\rho = 0.4$, $M < 3$, $\max_i H_i^- < 2$, and $\bar{v}_{\text{used}} < 0.285$ imply $\widehat{M} < 3$, $\max_i \widehat{H}_i^- < 2$, and $\widehat{v}_{\text{used}} < 0.7125$. These are *within-model* resources. Comparing them with observed market depth, accessible inventory, and traded volume would require asset-specific empirical calibration rather than equating D_F with a quoted order-book or ADV statistic.

Turnover and quadratic dissipation. The coordinate-distance bounds of Theorem 5.1 can be complemented by an aggregate estimate for constant per-share impact. Define

$$\mathcal{V}_2(t) = \frac{1}{2}\sum_{i=1}^{N}\big(S_i(t)^2 - F^2\big).$$

Proposition 5.2 (Turnover–dissipation bound under constant impact). *Suppose $g(S) \equiv \eta > 0$ and $S_i(0) = F$. Let the paired-dollar controls be absolutely continuous and assume that the entire price path satisfies $S_i(t) \geq a > 0$ and $|S_i(t) - F| \leq b$ for all i and $t \in [0,T]$. With $\dot{M}(t) = \sum_e \dot{z}_e(t)$ and $\mathrm{e} = \exp(1)$, the quadratic potential obeys the exact identity*

$$-\mathcal{V}_2(T) = \frac{\rho}{2}\int_0^T e^{-\rho(T-t)} \sum_{i=1}^{N} (S_i(t) - F)^2 \,\mathrm{d}t \geq 0 \tag{5.4}$$

and the turnover inequality

$$-\mathcal{V}_2(T) \leq \frac{\rho\eta b}{a}\int_0^T (T-u)e^{-\rho(T-u)}\dot{M}(u)\,\mathrm{d}u \leq \frac{\eta b}{a\,\mathrm{e}} M(T). \tag{5.5}$$

Proof. Set $x_i = S_i - F$. The price equation $\dot{S}_i = -\rho x_i + \eta \dot{X}_i / S_i$ and dollar pairing $\sum_i \dot{X}_i = 0$ give

$$\dot{\mathcal{V}}_2 = -\rho\mathcal{V}_2 - \frac{\rho}{2}\sum_i x_i^2, \qquad \mathcal{V}_2(0) = 0.$$

Variation of constants proves (5.4). Applying it to the level-price equation and using $S_i \geq a$ yields

$$\begin{aligned} \sum_i |x_i(t)| &\leq \frac{\eta}{a}\int_0^t e^{-\rho(t-u)} \sum_i |\dot{X}_i(u)|\,\mathrm{d}u \\ &\leq \frac{2\eta}{a}\int_0^t e^{-\rho(t-u)}\dot{M}(u)\,\mathrm{d}u. \end{aligned}$$

The second inequality follows from $\sum_i |\dot{X}_i| \leq 2\sum_e \dot{z}_e = 2\dot{M}$; oppositely directed flows at an asset can cancel in the net dollar rate. Since $\sum_i x_i^2 \leq b\sum_i |x_i|$, substitution into (5.4) and interchange of the nonnegative integrals give the first inequality in (5.5). The second follows from $\sup_{s\geq 0} \rho s e^{-\rho s} = \mathrm{e}^{-1}$. ∎

The time weight $\rho(T-u)e^{-\rho(T-u)}$ in (5.5) vanishes at $u=T$ and reaches its unique maximum at $T-u=1/\rho$; it also decays for trades made far before the terminal date. This describes the contribution of a dollar of turnover to the *upper-bound estimate*: late trades allow little time for recovery dissipation, while earlier disturbances decay before T. It does not identify an optimal trading schedule.

Two-session rolling-band example. Consider a uniform symmetric $\pm 25\%$ price band during each of two sessions, with the second session referenced to the preceding close. Then every price lies in $[9F/16, 25F/16]$, so $a=b=9F/16$ is admissible and

$$-\mathcal{V}_2(T) \le \frac{\eta}{\mathrm{e}} M(T).$$

If all N assets close at or below $(1-d)F$, $0<d<1$, then $-\mathcal{V}_2(T) \ge NF^2(2d-d^2)/2$, implying

$$M(T) \ge \frac{\mathrm{e}\, NF^2}{2\eta}(2d-d^2).$$

At $N=10$, $F=\eta=1$ and $d=0.02$, this becomes $M(T)\ge 0.5382$, compared with $M(T)\ge 0.198$ from Theorem 5.1. Both bounds remain below $M=3$.

For partial coverage, appreciation of the compensating assets can offset part of the falling subset's potential deficit. The aggregate potential itself cannot become positive by (5.4); without further constraints on compensating prices, it need not provide a useful positive lower bound on turnover for a specified falling subset alone.

6. Finite trading rates and abstract price bands.

A structurally reachable loss may require trades that cannot be completed at finite speed. We therefore restrict the control set by gross executed-share rates and single-session price bands. These restrictions preserve the barrier theorems, while positive reachability requires a new continuous construction rather than the pulse controls used in Market C.

The imposition of finite execution speed builds on rate-constrained trading models such as Yang [26], while our goal remains the reachability of a collective loss threshold rather than the optimum of an agent's execution objective.

6.1. Two separate execution restrictions.

For absolutely continuous nonnegative paired-dollar edge rates $\dot z_e(t)$, define an asset's *gross executed share rate* by

$$r_i^{\mathrm{gross}}(t) = \frac{1}{S_i(t)} \sum_e |B_{ie}| \dot z_e(t), \qquad 0 \le r_i^{\mathrm{gross}}(t) \le \bar v, \qquad i=1,\dots,N. \tag{6.1}$$

This bound counts buy and sell executions *before* netting. In particular $|\dot\theta_i| \le r_i^{\mathrm{gross}} \le \bar v$ and instantaneous pulses are inadmissible. Bounding only the signed net share rate would allow simultaneous offsetting gross transactions of arbitrary size and would not constrain executed turnover.

Independently, an abstract *one-session price-band constraint* restricts every executed price to

$$F(1-\beta_-) \le S_i(t) \le F(1+\beta_+), \qquad 0<\beta_-<1, \quad \beta_+>0. \tag{6.2}$$

The control must keep its executed-price path inside this interval; a control that would leave the band is *inadmissible*, not reflected back into the band. This is a prescribed mathematical state constraint, not a reconstruction of any named exchange's trading rules. In particular, it has no exchange-specific tick sizes, price-tier tables, order queues, auction mechanisms, or conditional band enlargement; actual exchange rules are discussed only as external context in Section 8.

Proposition 6.1 (Rate and price-band exclusion certificates). *Suppose the common impact function is positive and $g(S) \le G$ on the whole admissible price region. Under the cap* (6.1)*, every asset obeys*

$$S_i(t) \ge F - \frac{G\bar{v}}{\rho}(1 - e^{-\rho t}), \qquad 0 \le t \le T. \tag{6.3}$$

Under the additional price band (6.2)*, put*

$$S_{\text{floor}} = \max\left\{F(1-\beta_-),\ F - \frac{G\bar{v}}{\rho}(1 - e^{-\rho T})\right\} > 0. \tag{6.4}$$

Then for any breadth α and any admissible control,

$$\boxed{\mathcal{C}_\alpha(t) \le \log \frac{F}{S_{\text{floor}}} \quad \text{for all } t \le T.} \tag{6.5}$$

In particular, a necessary finite-speed condition to reach threshold $\delta > 0$ is

$$\boxed{\bar{v} \ge \frac{\rho F(1 - e^{-\delta})}{G(1 - e^{-\rho T})}.} \tag{6.6}$$

A one-session lower price limit independently excludes any $\delta > -\log(1-\beta_-)$. These are necessary *constraints, generally not sharp reachability thresholds. For comparison, if a* hypothetical *multi-session regime resets a proportional lower band to each preceding close, then over D sessions the elementary bound is only $S_i(T_D) \ge F\prod_{d=1}^{D}(1-\beta_{-,d})$ and $\mathcal{C}_\alpha(T_D) \le -\sum_{d=1}^{D}\log(1-\beta_{-,d})$. The cumulative ceiling depends on the successive closing-price resets.*

Proof. The variation-of-constants formula for $\dot{S}_i = -\rho(S_i - F) + g(S_i)\dot{\theta}_i$ and $|\dot{\theta}_i| \le \bar{v}$ yields $S_i(t) \ge F - G\bar{v}\int_0^t e^{-\rho(t-s)}\,\mathrm{d}s$, proving (6.3). The price band supplies the second floor; every individual log loss is at most $\log(F/S_{\text{floor}})$, and so are its cross-sectional mean, upper-breadth threshold, and their minimum. Rearranging the rate-only floor is necessary for an individual loss to reach δ. ∎

Dollar pairing and terminal closure yield additional market-wide execution requirements. Since each paired dollar is counted twice in gross share execution,

$$M = \frac{1}{2}\int_0^T \sum_i S_i(t) r_i^{\text{gross}}(t)\,\mathrm{d}t \le \frac{NF(1+\beta_+)\bar{v}T}{2}. \tag{6.7}$$

Together with Theorem 5.1, this gives the necessary joint constraint

$$\boxed{\bar{v} \geq \frac{2k}{NF(1+\beta_+)T} \int_{Fe^{-\delta}}^{F} \frac{u}{g(u)}\, \mathrm{d}u.} \tag{6.8}$$

For a closed executed-dollar ledger, the nonnegative cost identity (2.11) also yields

$$E(S_i(T)) \leq F\bar{v}T, \qquad \bar{v} \geq \frac{E(Fe^{-\delta})}{FT} \quad \text{for every qualifying asset.} \tag{6.9}$$

Bounds (6.6), (6.8), and (6.9) are distinct necessary tests; satisfying them does not construct a crisis path.

A per-asset rate estimate gives a first-order $O(\bar{v})$ bound on systemic losses. Under dollar pairing, the leading-order price movements cancel in the market mean, yielding the quadratic small-rate bound below for constant impact.

Theorem 6.2 (A quadratic small-rate crisis barrier for constant impact). *Let $g(S) \equiv \eta > 0$, assume dollar pairing $\sum_i S_i\dot{\theta}_i = 0$, and impose the gross share-rate cap (6.1). Write $d_T = 1 - e^{-\rho T}$ and*

$$b_T = \frac{\eta\bar{v}}{F\rho}\, d_T < 1. \tag{6.10}$$

Then, with or without terminal dollar closure and for every *$N \geq 2$ and breadth α,*

$$\boxed{\mathcal{C}_\alpha(T) \leq \left(\frac{\eta\bar{v}}{F\rho}\right)^2 \left[d_T - \rho T e^{-\rho T} + \frac{d_T^2}{2(1-b_T)^2}\right].} \tag{6.11}$$

Consequently, the supremum of attainable systemic losses is $O(\bar{v}^2)$ as $\bar{v} \downarrow 0$. For fixed N, T and positive price-band widths on both sides of the benchmark, Theorem 6.3 supplies a matching construction. Here $\bar{v}_*(\delta)$ denotes the infimum rate cap permitting severity at least δ, with fixed $N, T, F, \rho, \eta, \alpha$, positive per-asset initial inventory allowances, terminal dollar closure, and prescribed bands $\beta_- > 0$, $\beta_+ > 0$. In this constrained class, $\bar{v}_*(\delta) = \Theta(\sqrt{\delta})$ as $\delta \downarrow 0$. These bounds determine the scaling order; the optimal constants remain undetermined.

Proof. Set $x_i(t) = S_i(t) - F$ and $m(t) = \sum_i x_i(t)$. Variation of constants gives $|x_i(t)| \leq \eta\bar{v}(1 - e^{-\rho t})/\rho$. Since $\sum_i S_i\dot{\theta}_i = 0$, $\sum_i \dot{\theta}_i = -F^{-1}\sum_i x_i\dot{\theta}_i$ and therefore $\dot{m} = -\rho m - (\eta/F)\sum_i x_i\dot{\theta}_i$. Integrating and bounding the gross rates yields

$$\frac{|m(T)|}{NF} \leq \left(\frac{\eta\bar{v}}{F\rho}\right)^2 [d_T - \rho T e^{-\rho T}]. \tag{6.12}$$

For $|x_i(T)|/F \leq b_T < 1$, Taylor's inequality $-\log(1+y) \leq -y + y^2/[2(1-b_T)^2]$ implies $\overline{L}(T) \leq |m(T)|/(NF) + b_T^2/[2(1-b_T)^2]$. Use $\mathcal{C}_\alpha \leq \overline{L}$ to obtain (6.11). The construction in Theorem 6.3 has $\mathcal{C}_1(T) = c_N a^2 + O(a^3)$ with $c_N > 0$, and its gross rate is $O(a)$; setting $a \propto \bar{v}$ gives a positive loss of matching order $\Omega(\bar{v}^2)$ while keeping a fixed positive band. This proves the stated small-rate exponent. ∎

Theorem 6.3 (Positive all-asset decline with bounded continuous execution). *Assume $g \in C^3$ near F, $g(F) > 0$, $\rho > 0$, and $e_0 = Fg'(F)/g(F) < 1$. Fix* any *$N \geq 2$, horizon $T > 0$, rate cap $\bar{v} > 0$, one-session band widths $\beta_-, \beta_+ > 0$, and any fixed positive amount of initially available shares per asset. Then there exists an* absolutely continuous, *paired-dollar, per-asset gross-rate-bounded control, remaining strictly inside the band, with $X_i(T) = 0$ for each asset and*

$$\boxed{S_i(T) < F \text{ for every } i, \qquad \mathcal{C}_1(T) > 0.} \tag{6.13}$$

The attained loss can be arbitrarily small. This does not *assert that a prescribed fixed large threshold remains reachable under an arbitrarily tight cap or band.*

Proof. Choose a zero-sum vector $u \in \mathbb{R}^N$ with every $u_i \neq 0$. Put $\Delta = T/3$, $r = e^{-\rho\Delta}$ and set the bounded, three-bin signal

$$f(t) = \begin{cases} 1, & 0 \leq t < \Delta, \\ -(1+r), & \Delta \leq t < 2\Delta, \\ r, & 2\Delta \leq t \leq T, \end{cases} \qquad \dot{X}_i = a u_i f(t). \tag{6.14}$$

The dollar rates sum to zero, and $\int_0^T f(t)\,\mathrm{d}t = 0$.

Direct integration also gives $\int_0^T e^{-\rho(T-t)} f(t)\,\mathrm{d}t = 0$. Write $h(S) = g(S)/S$ and expand $S_i(t) = F + a u_i y(t) + a^2 u_i^2 z(t) + O(a^3)$, where $\dot{y} = -\rho y + h(F) f$, $y(0) = 0$, and $\dot{z} = -\rho z + h'(F) f y$, $z(0) = 0$. The weighted cancellation implies $y(T) = 0$. Since $h'(F)/h(F) = (e_0 - 1)/F$, integration by parts yields

$$\boxed{\log \frac{F}{S_i(T)} = \frac{\rho(1-e_0)}{2F^2}\, a^2 u_i^2 \int_0^T e^{-\rho(T-t)} y(t)^2\,\mathrm{d}t + O(a^3).} \tag{6.15}$$

The integral is strictly positive. Thus all prices fall for every sufficiently small $a > 0$. Represent each zero-sum dollar-rate vector by nonnegative source-to-destination edge rates with minimum gross flow, so $r_i^{\text{gross}} = |a u_i f|/S_i$. The entire price path converges uniformly to F as $a \downarrow 0$, ensuring compliance with the rate cap, price band, and any positive inventory allowance for sufficiently small a. ∎

The continuous construction establishes local common-loss reachability under a finite rate cap and terminal dollar closure.

Interaction of rate and price-band constraints. The global barrier for $e(S) \geq 2$ and the full-coverage exclusion for $e(S) \geq 1$ remain valid after restricting the set of admissible controls. For $p = 1$, a bounded two-stage buy-then-sell dollar-rate control still makes $k < N$ assets end below F: its per-asset signed dollars integrate to zero, while $\sum_i S_i(T) = NF$ and strict concavity of log makes the average log loss positive. In contrast, the *arbitrarily severe* construction for $p = 0$ in Theorem 4.1(C) uses instantaneous sweeps and therefore falls outside the rate-constrained class (6.1). Under a fixed price band, no one-session terminal loss can exceed $-\log(1-\beta_-)$. Theorem 6.3 establishes strictly positive local all-asset declines under both restrictions, with severity limited by the rate and price-band constraints.

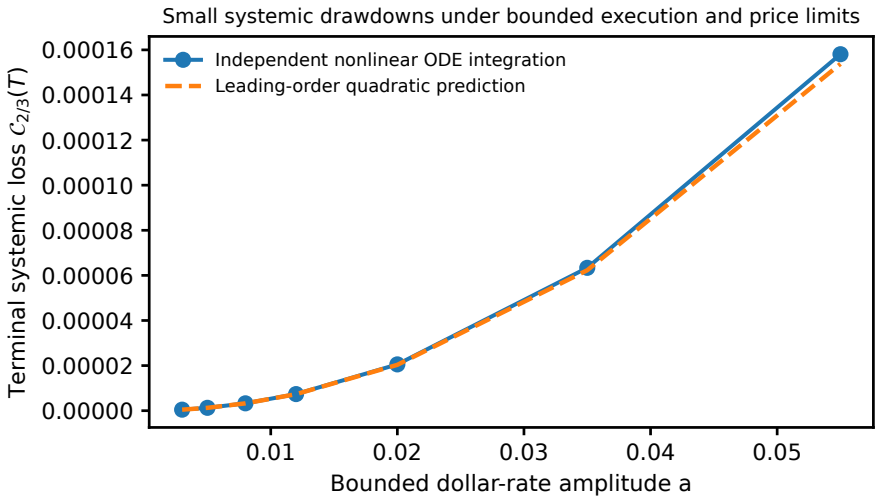


Figure 3. ***Quadratic local all-asset loss under a finite-rate control.*** *Independently integrated nonlinear price paths for constant g agree with the leading-order quadratic prediction from* (6.15) *as the amplitude decreases. The displayed functional is $C_{2/3}(T)$, evaluated for the all-asset-decline construction of Theorem 6.3. The plotted amplitude is sufficiently small to preserve positive prices; the admissibility checks for the highlighted $a = 0.035$ case are reported in the code bundle.*

Table 1

How execution restrictions affect reachability for $p = 0$. Arbitrarily severe partial-coverage losses use a sufficiently long fixed finite horizon, as in Theorem 4.1(C). Local reachability holds for every $T > 0$; "local" refers to a sufficiently small positive terminal all-asset loss.

Admissible controls	Established reachability	Scope of the result
Instantaneous sweeps, no band	Arbitrarily severe partial-coverage log loss, $k < N$	Does not model a bounded execution rate
Gross rate $\bar{v}$ only	Local all-asset loss for any $\bar{v} > 0$	Unbounded severe-loss construction not certified
Hard session band only	Local all-asset loss when $\beta_- > 0$ and $\beta_+ > 0$	$\delta > -\log(1-\beta_-)$ impossible
Both restrictions	Local all-asset loss for positive caps/widths; rate and price-floor certificates	No general exact severe threshold or optimal execution rate

Figure 3 compares the numerical small-rate loss with its leading-order quadratic prediction.

7. Systemic drawdowns under rolling price-band constraints. This section studies a *hypothetical proportional-reset price-band protocol*, not a Tokyo Stock Exchange case study or an attempt to replicate any other venue. A price band restricts *executed* prices during each modeled session and resets from the previous modeled closing price. The multiday analysis tracks the assets repeatedly closing at the lower band, their contribution to the crisis functional, and the capacity of the remaining assets to absorb paired dollar flow. Feasibility requires sufficient breadth and average loss, selling speed, and receiving capacity within the prescribed

rate caps and bands.

A *lower-band close* (called a *lower-limit close* in some theorem statements) means only that the last executed price equals the *modeled* session floor. Neither such a closing print nor repeated floor hits imply a real-exchange limit-down event, a locked order book, an unfilled sell queue, or a market halt; those would require additional institutional rules and state variables. This mathematical convention applies equally to the numerical certificate below. Empirical exchange price-limit evidence is discussed separately in Section 8.

7.1. Rolling bands and the exact cumulative accounting. Partition trading time into D sessions $[T_{d-1}, T_d]$, each of duration $\tau_d > 0$. Overnight calendar time does not contribute to the model clock; prices carry over continuously from the preceding close. In the stylized *proportional-reset* version, set $R_{i,d} = S_i(T_{d-1})$ and require

$$(1-\beta_{i,d,-})R_{i,d} \le S_i(t) \le (1+\beta_{i,d,+})R_{i,d}, \qquad t \in [T_{d-1}, T_d], \tag{7.1}$$

where $0 < \beta_{i,d,-} < 1$. A lower-limit close is $S_i(T_d) = (1-\beta_{i,d,-})R_{i,d}$; a D-day streak means this equality holds for the *same* assets in D consecutive sessions. A streak in $k = \lceil \alpha N \rceil$ assets is not yet a systemic crisis under Definition 2.2: the market-average log loss must also reach the chosen threshold.

For a common constant lower fraction $q = 1 - \beta_-$, each streaking asset has $S_i(T_D) = Fq^D$ and loss $L_i(T_D) = D\lambda$ with $\lambda = -\log q$. Conversely, even without a streak, the state constraint ensures

$$\mathcal{C}_\alpha(T_D) \le D\lambda \tag{7.2}$$

for a common band and initial equal benchmark price. For heterogeneous bands the right side can be replaced by $\max_i \sum_{d=1}^{D} [-\log(1-\beta_{i,d,-})]$. These are *geometric ceilings*, not sufficient conditions for a dollar-paired trajectory to hit the floor.

All subsequent multiday statements and the computer-assisted example use only the proportional-reset constraint (7.1). No price-tiered yen rule, exchange-specific closing auction, or conditional enlargement event is imposed.

7.2. Cross-sectional crisis threshold under repeated lower-band closes. Consecutive lower-limit closes by k stocks satisfy the breadth requirement, but do not automatically satisfy the mean-loss requirement: the remaining assets may appreciate as they receive matched dollars. This is the same $k < N$ compensation mechanism that permits Market C's partial-coverage outcomes, now seen from the opposite direction. The following exact endpoint identity shows when the falling majority becomes a crisis under the *unchanged* functional, without yet constructing the trades.

Proposition 7.1 (A conditional systemic-crisis trigger test). *Suppose the same $k = \lceil \alpha N \rceil < N$ assets close at Fq^D after D identical proportional-reset sessions, with $q \in (0,1)$. Let the remaining $N-k$ assets close at $Y_j > 0$, and write $\lambda = -\log q$. For any $0 < \delta \le D\lambda$, the* terminal *crisis threshold $\mathcal{C}_\alpha(T_D) \ge \delta$ holds if and only if*

$$\boxed{\prod_{j=1}^{N-k} \frac{Y_j}{F} \le \exp\bigl(kD\lambda - N\delta\bigr).} \tag{7.3}$$

If all other assets have individual log losses at most $D\lambda$ (or if $k > N/2$, so the $N-k$ other assets cannot fill the top k ranks), then $Q_\alpha(T_D) = D\lambda$ and the terminal crisis severity is exactly

$$\boxed{\mathcal{C}_\alpha(T_D) = \min\left\{ D\lambda,\ \frac{kD\lambda - \sum_{j=1}^{N-k} \log(Y_j/F)}{N} \right\}.} \tag{7.4}$$

In particular, with a single compensator ($k = N-1$), its terminal price must obey $Y/F \le \exp(kD\lambda - N\delta)$ if a crisis of threshold δ is to occur. These are exact conditional endpoint identities*, not sufficient conditions for a trade path realizing the stipulated closing prices.*

Proof. The k streaking assets have loss $D\lambda$, hence $Q_\alpha \ge D\lambda$. For $\delta \le D\lambda$, the coverage part of the crisis criterion is therefore automatic, and the remaining requirement is $\overline{L}(T_D) \ge \delta$. Substituting the terminal prices into $\overline{L}(T_D) = N^{-1}\sum_i \log[F/S_i(T_D)]$ gives (7.3). If every other loss is at most $D\lambda$, or if $N-k<k$, the kth-largest loss is exactly $D\lambda$, proving (7.4). ∎

For the interval-certified $N = 10$, $k = 9$, $q = 0.97$, $D = 2$ construction of Proposition 7.6, $D\lambda \approx 0.060918$. The compensator closes at $Y/F \approx 1.405161$ in the high-accuracy reconstruction, yielding $\mathcal{C}_{0.9} \approx 0.020811$. The outward-rounded certificate gives $1.4047874 < Y/F < 1.4055543$, which rigorously implies $\mathcal{C}_{0.9} > 0.020$ for the specified trajectory. The threshold $\delta = 0.020$ requires $Y/F \le \exp(9D\lambda - 10(0.020)) \approx 1.416609$, which the certified path satisfies. Thus the receiving asset's terminal price determines whether the same broad lower-limit streak qualifies as a systemic loss event.

Figure 4 illustrates the distinction between repeated closing-limit losses and the systemic crisis criterion.

7.3. A minimum speed for consecutive lower-band closes. A closing price at the lower limit must be reached in finite time despite recovery toward F. For $g(S) \equiv \eta$, start a session at price s and suppose its length is τ and its lower floor is $(1-\beta)s$. Even before asking who purchases the matched dollars, the gross share-rate cap bounds the fastest decline through $\dot z = -\rho(z-F) - \eta\bar v$, $z(0) = s$. This yields a necessary threshold but leaves the market-wide matching problem unresolved.

Proposition 7.2 (Rolling-limit shock-rate certificates). *In the constant per-share impact model, a lower-limit close in a session starting at $s > 0$ requires*

$$\boxed{\bar v \ge \left[\frac{\rho}{\eta}\left(F - s + \frac{\beta s}{1 - e^{-\rho\tau}}\right)\right]_+.} \tag{7.5}$$

If the same asset closes at the same proportional lower limit on each of D sessions of common duration τ, with $s_d = Fq^{d-1}$, then

$$\boxed{\bar v \ge \frac{\rho F}{\eta} \max_{1\le d\le D}\left[1 + q^{d-1}\left(\frac{1-q}{1-e^{-\rho\tau}} - 1\right)\right].} \tag{7.6}$$

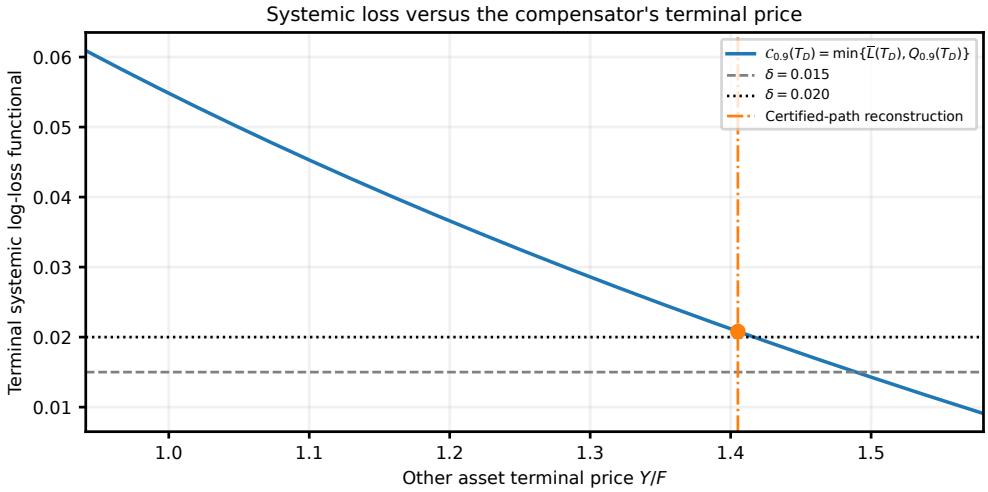


Figure 4. ***Systemic loss and the compensating asset's terminal price.*** *Conditional terminal severity for $N = 10$, nine assets closing twice at a 3% daily lower limit, versus the remaining asset's terminal price. The dotted marker denotes the constructed dollar-closed path; the reference lines show the log-loss thresholds $\delta = 0.015$ and $\delta = 0.020$. The curve gives the conditional algebraic relation between the terminal price and systemic loss.*

Moreover, when $\eta\bar{v} < \rho F$, all prices starting at F obey the perpetual rate-only *floor $S_i(t) \geq F - \eta\bar{v}/\rho$, implying the necessary integer-streak bound*

$$\boxed{D \leq \frac{\log(1 - \eta\bar{v}/(\rho F))}{\log q}.} \tag{7.7}$$

The bound is understood only for $\eta\bar{v} < \rho F$; with a larger rate cap it supplies no finite-day obstruction. None of these scalar inequalities is sufficient for multi-asset price-band feasibility or terminal dollar closure.

Proof. Variation of constants and $\dot{\theta}_i \geq -\bar{v}$ imply $S_i(\tau) \geq F + (s - F)e^{-\rho\tau} - (\eta\bar{v}/\rho)(1 - e^{-\rho\tau})$. Requiring $S_i(\tau) \leq (1 - \beta)s$ yields (7.5); substitute $s = Fq^{d-1}$ for each day to obtain (7.6). The same scalar comparison from initial F gives $S_i(t) \geq F - (\eta\bar{v}/\rho)(1 - e^{-\rho t})$, hence the uniform floor; combine it with Fq^D and note $\log q < 0$ to deduce (7.7). ∎

7.4. Obstruction to universal consecutive lower-band closes. The all-asset case is especially restrictive: there is no complement capable of absorbing a net dollar flow. The same potential that gave the original exclusion barriers now rules out a second simultaneous decline under a simple below-benchmark band condition. This obstruction extends the breadth distinction in the A/B/C regimes to the entire market.

Theorem 7.3 (An obstruction to a second all-asset limit-down close). *Assume $g_i(S) > 0$, $\rho_i > 0$, unchanged fundamental benchmarks $F_i > 0$, and instantaneous interasset executed-dollar balance $\sum_i \mathrm{d}X_i = 0$. Suppose every asset closes the first session at $S_i(T_1) = q_i F_i < F_i$, and the* entire second-session admissible band *lies at or below its own benchmark,*

$$(1 + \beta_{i,2,+})q_i \leq 1 \quad \textit{for every } i. \tag{7.8}$$

Then it is impossible for every *asset to close the second session strictly below its first close. In particular, simultaneous second-day lower-limit closes in all N assets are impossible. This holds regardless of the transaction-rate cap, terminal signed-dollar balances, or the functional form of the positive g_i.*

Proof. Use $\Phi_i'(S) = S/g_i(S) > 0$ and $H = \sum_i \Phi_i(S_i)$. Trade settlement preserves H, and recovery contributes $\dot{H} = -\sum_i \rho_i S_i(S_i - F_i)/g_i(S_i) \geq 0$ whenever every $S_i \leq F_i$. Condition (7.8) keeps the full second-session price path in that region. Hence $H(T_2) \geq H(T_1)$. If every second close were below the preceding close, strict monotonicity of all Φ_i would instead give $H(T_2) < H(T_1)$, a contradiction. ∎

For the stylized symmetric proportional band, $q_i = 1 - \beta_i$ and $\beta_{i,2,+} = \beta_i > 0$, so $(1 + \beta_i)(1 - \beta_i) = 1 - \beta_i^2 < 1$ and the theorem applies. The theorem uses full execution, paired-dollar settlement, and fixed fundamentals. Extensions with unmatched orders, an external cash account, or changing fundamentals require separate analysis. It leaves open consecutive streaks in $k < N$ assets supported by offsetting dollar flows in the remaining assets.

The same model cannot represent a positive-duration *all-asset locked limit-down state* with every price fixed strictly below F_i: maintaining $\dot{S}_i = 0$ would require $\dot{X}_i = \rho_i S_i(S_i - F_i)/g_i(S_i) < 0$ for every asset, contradicting $\sum_i \dot{X}_i = 0$. A limit-hit closing print is instead permitted as an endpoint condition. Persistent unmatched sell queues require a new order-book state, not simply an extended price trajectory.

7.5. A positive multi-day resource and accounting constraint. Even when not all assets fall, each target that is below F faces upward recovery. Terminal dollar closure therefore requires a path-history resource, not just a final selling flow: earlier above-benchmark excursions must provide the positive recovery contribution that compensates later losses. This is another expression of the same potential introduced in Section 3, now integrated over successive trading sessions.

Proposition 7.4 (Preconditioning needed for a dollar-closed limit-down streak). *Let an asset begin at F and finish D sessions at $s_D < F$, with $X_i(T_D) = 0$. Then its trajectory necessarily visits $S_i > F$ on a set of positive trading-time measure and satisfies*

$$\rho \int_0^{T_D} \frac{S_i(t)(S_i(t) - F)}{g(S_i(t))}\,\mathrm{d}t = \int_{s_D}^{F} \frac{u}{g(u)}\,\mathrm{d}u > 0. \tag{7.9}$$

If every session after the first keeps the asset at or below F, and the first-session upper band is $U_1 > F$, a necessary condition is

$$\boxed{\rho\tau_1 \max_{F \leq u \leq U_1} \frac{u(u - F)}{g(u)} \geq \int_{s_D}^{F} \frac{u}{g(u)}\,\mathrm{d}u.} \tag{7.10}$$

For constant impact $g = \eta$, and $s_D = Fq^D$, $U_1 = F(1 + \beta_+)$, this simplifies to $\rho\tau_1 \geq (1 - q^{2D})/[2\beta_+(1 + \beta_+)]$. Writing $H_+ = \rho\tau_1\beta_+(1 + \beta_+)$, whenever $2H_+ < 1$ this imposes the additional necessary streak-length bound

$$\boxed{D \leq \frac{\log(1 - 2H_+)}{2\log q}.} \tag{7.11}$$

This bound is specific to terminal dollar closure and the absence of above-fundamental excursions in later sessions. It is not sufficient to produce a consecutive limit-down path.

Proof. Integrate the identity $\mathrm{d}\Phi(S_i) = \mathrm{d}X_i - \rho S_i(S_i - F)\,\mathrm{d}t/g(S_i)$ over the full horizon. Since $X_i(T_D) = 0$ and $\Phi(F) = 0$, the integral must equal $-\Phi(s_D) > 0$. Any portion of the trajectory at $S_i \le F$ contributes a nonpositive integrand, so the positive part must arise above F. If later sessions cannot rise above F, all positive contribution must be earned in session 1, whose supremum is bounded by the left side of (7.10). ∎

7.6. A necessary receiving-capacity bound for multiasset execution. For $k < N$, a subset's net executed-dollar outflow must be absorbed by the remaining assets. Partition the assets into a falling subset J and its complement J^c. On a complete trading graph with node-level gross-rate caps and no separate edge capacities, conservation and the receiving assets' prices and rates bound the dollar flow across this partition.

Proposition 7.5 (A market-cut capacity and consecutive-limit lower bound). *Fix a subset J of assets and write J^c for its complement. Under paired-dollar execution and the per-asset gross share-rate cap $\bar{v}$, every admissible absolutely continuous path satisfies, for almost every t,*

$$\boxed{\left|\sum_{i\in J}\dot{X}_i(t)\right| \le \bar{v}\min\left\{\sum_{i\in J}S_i(t),\sum_{j\in J^c}S_j(t)\right\}.} \tag{7.12}$$

Suppose that throughout a session $[T_{d-1}, T_d]$ of length τ_d, each $i \in J$ stays at or below its unchanged benchmark F_i and ends at $s_{i,d} < s_{i,d-1}$. Let $\Phi_i'(S) = S/g_i(S) > 0$ and put

$$K_{J,d} := \sum_{i\in J}\left[\Phi_i(s_{i,d-1}) - \Phi_i(s_{i,d})\right] > 0. \tag{7.13}$$

If all compensating assets satisfy $S_j(t) \le U_{j,d}$ during that session, then the session's paired executed-dollar turnover M_d and the required gross-share-rate cap satisfy

$$\boxed{M_d \ge K_{J,d}, \qquad \bar{v} \ge \frac{K_{J,d}}{\tau_d \sum_{j\in J^c} U_{j,d}}.} \tag{7.14}$$

When $J^c = \varnothing$, the hypothesized simultaneous strict decrease of all assets cannot occur on a session lying entirely below benchmarks.

Proof. Dollar pairing gives $\sum_{i\in J}\dot{X}_i = -\sum_{j\in J^c}\dot{X}_j$. In either set, $|\dot{X}_i| \le S_i r_i^{\mathrm{gross}} \le S_i\bar{v}$, proving (7.12). The price-response coordinate obeys $\Delta X_i = \Phi_i(s_{i,d}) - \Phi_i(s_{i,d-1}) + \rho_i \int_{T_{d-1}}^{T_d} S_i(S_i - F_i)/g_i(S_i)\,\mathrm{d}t$. The integral is nonpositive for $i \in J$, hence $\sum_{i\in J}\Delta X_i \le -K_{J,d}$. The net executed dollars entering J^c are therefore at least $K_{J,d}$. The total matched-dollar turnover must cover this flow, and the complement's gross-share capacity is no more than $\bar{v}\tau_d \sum_{j\in J^c} U_{j,d}$. These yield (7.14). If J^c is empty, there is no opposing flow. ∎

A nonempty complement can impose an additional rate requirement through its receiving capacity. Its appreciation also offsets the market-average log loss in Proposition 7.1, so feasibility depends on both dollar absorption and the resulting average loss.

For constant per-share impact $g_i \equiv \eta$, $k = |J|$ identical lower-limit assets, initial benchmark F, common lower-limit ratio $q \in (0,1)$, and $m = N-k$ compensators with common upper-band fraction $\beta_{c,+}$, the dth-day specialization is

$$\boxed{\bar{v} \geq \frac{k}{m} \frac{F\left(q^{2(d-1)} - q^{2d}\right)}{2\eta\tau_d(1+\beta_{c,+})^d} \quad (d \geq 2),} \tag{7.15}$$

provided all k target prices remain below F and each compensator's preceding reference price has evolved from F under the specified upper bands. The k/m multiplier makes the funding bottleneck explicit. The condition gives a *necessary* lower bound on the rate. For the ten-asset, two-day example ($k = 9, m = 1, q = 0.97, \beta_{c,+} = 0.25, \tau_d = 20$), it gives only $\bar{v} \gtrsim 0.0080$; the stronger systemic quadratic bound below is more informative for that particular configuration.

7.7. A certified two-session construction and bounds on the critical rate. We construct and certify a path satisfying all the preceding constraints. Fix $N = 10$, with nine identical target assets and one compensator, $F = \eta = 1$, $\rho = 0.4$, two sessions of length 20, target rolling bands $\pm 3\%$, compensator bands $\pm 25\%$, and terminal closure $X_i(40) = 0$ for every asset. The targets share one prescribed continuous price path $x(t)$; the compensator follows the scalar nonlinear ODE determined by paired dollars. This symmetry reduces the calculation to two price states and establishes feasibility for the specified ten-asset model. A reproducible fixed-point interval calculation certifies the nonlinear compensator dynamics and all execution constraints for the piecewise-linear target path.

Proposition 7.6 (A computer-assisted two-day rate certificate). *In the ten-asset specification above ($k = 9$, $F = \eta = 1$, $\rho = 0.4$, $\tau_1 = \tau_2 = 20$, target bands $\pm 3\%$, compensator bands $\pm 25\%$), there exists an absolutely continuous, fully paired-dollar path with all per-asset terminal signed-dollar ledgers exactly zero such that the same nine assets close at 0.97 and 0.9409, respectively, while*

$$\boxed{\mathcal{C}_{0.9}(40) > 0.020, \quad \bar{v}_{\text{used}} < 0.285, \quad M < 3, \quad H_i^- < 2 \quad (1 \leq i \leq 10).} \tag{7.16}$$

Every executed price remains inside its rolling session band. Here M counts paired executed dollars once and H_i^- denotes gross shares sold, so initial holdings of two shares of every asset are sufficient without shorting.

Proof sketch. Let $r = 23/10$. Prescribe the common target price $x(t)$ as the continuous piecewise-linear interpolation of

$$(t,x) = (0,1), (r,h), (20-r,h), (20,0.97), (20+r,0.995), (40-r,0.995), (40,0.9409). \tag{7.17}$$

The first plateau h is the unique root near 1.02 of the exact net-dollar-closure equation

$$\begin{aligned} &101600000000h^2 - 97138000000h - 6646157699 = 0, \\ &1.02020225507905 < h < 1.02020225507908. \end{aligned} \tag{7.18}$$

The compensator $y(t)$ is the unique positive solution of $\dot{y} = -0.4(y-1) - 9u(t)/y$, $y(0) = 1$, where $u(t) = x(t)[\dot{x}(t) + 0.4(x(t)-1)]$. All ten executed-dollar flows sum to zero at each time,

and the exact root equation gives $X_i(40) = 0$ for every target, hence also for the compensator. A validated interval integration gives

$$\begin{gathered} 1.2219388 < y(20) < 1.2225356, \quad 1.4047874 < y(40) < 1.4055543, \\ \sup_t 9|u(t)|/y(t) < 0.284481, \\ \int_0^{40} 9[u(t)]_+/y(t)\,\mathrm{d}t < 1.880, \qquad 9\int_0^{40} |u(t)|\,\mathrm{d}t < 2.984. \end{gathered} \tag{7.19}$$

The targets have even smaller gross rates and gross selling shares; sessionwise interval enclosures verify both rolling bands. Exact rational enclosures for $e^{0.2}$ verify $\overline{L}(40) > 0.020$, while the nine target losses are $-\log(0.9409) > 0.020$, proving (7.16). ∎

For this *fixed* ten-asset, two-day, nine-streaking-assets instance, let $\bar{v}_*$ be the infimum gross rate cap among *all* admissible paths also satisfying $\mathcal{C}_{0.9}(40) \geq 0.020$. The quadratic system-wide bound (6.11) implies for every $\bar{v} \leq 0.044$ (with $F = \eta = 1$, $\rho = 0.4$, $T = 40$):

$$\mathcal{C}_{0.9}(40) \leq (\bar{v}/0.4)^2 \left[1 + \frac{1}{2(1-\bar{v}/0.4)^2}\right] \leq \frac{1563441}{79210000} < 0.020.$$

Since the right-hand side of (6.11) is continuous in $\bar{v}$ near 0.044 and is strictly below 0.020 at that value, the exclusion extends to some rates above 0.044. Together with the feasible construction in Proposition 7.6, this yields the rigorous, though nonoptimal, bracket for the critical gross-rate cap:

$$\boxed{0.044 < \bar{v}_* < 0.285.} \tag{7.20}$$

The certified trajectory supplies a feasible upper rate, while the quadratic barrier gives a strict lower bound for the same market instance. Determining the exact critical rate and whether it is attained remains a constrained multiday optimal-control problem.

Sensitivity and the role of symmetry. The ten-asset certificate deliberately reduces nine identical target states to one trajectory. Holding that *same* $x(t)$ and exact target-dollar closure fixed, a numerical sensitivity check repartitions ten assets into k identical targets and $10 - k$ identical compensators. Their common compensator price obeys $\dot{y} = -0.4(y-1) - [k/(10-k)]u(t)/y$. Table 2 shows why the number of receiving assets matters: the specified control achieves the fixed 0.020 mean-loss target only for $k = 9$ among the cases tested. Changing k also changes the coverage fraction $k/10$; it is not a test at fixed $\alpha = 0.9$. For $k < 9$, the rows are high-accuracy ODE diagnostics for this fixed control; feasibility under redesigned controls remains open. The apparent sensitivity of the $k = 9$ case to a wide compensator band and the $k/(10-k)$ receiving-capacity burden are features of the chosen witness, not universal market thresholds. For $k = 9$, the feasibility bounds of Proposition 7.6 are interval-certified. Similarly, for the fixed $k = 9$ path, the certificate proves compliance at a symmetric compensator band of $\pm 24.7\%$ and proves violation at $\pm 24\%$ (the price interval at $t = 17.7$ lies strictly below 0.76). Changing the band does not rerun or optimize the control.

Table 2

Fixed-control sensitivity to cross-sectional composition. The tabulated prices, mean log losses, and band widths are high-precision numerical reconstructions. For $k = 9$, feasibility under the stated constraints is additionally established by the interval certificate; Section 7.7 gives the certified bounds. The reported band width refers to the fixed path, without optimizing the control.

Targets k	Compensators	$y(40)$	Mean log loss	Computed band for fixed path
6	4	1.08112	0.00535	4.28%
7	3	1.12302	0.00784	6.55%
8	2	1.20141	0.01204	10.88%
9	1	1.40516	0.02081	24.55%

Figure 5 displays the price paths of the certified construction.

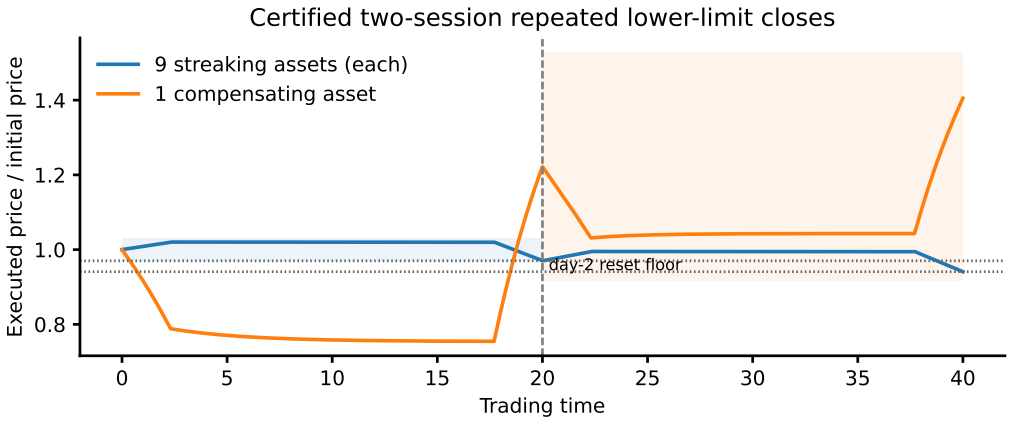


Figure 5. ***An interval-certified two-session path.*** *Continuous target and compensator prices for the construction in Proposition 7.6. The nine targets close at the reset lower limits; the compensator stays inside its 25% daily limits. Curves are illustrative numerical reconstructions; admissibility is independently certified by the outward-rounded arithmetic described in Appendix G.*

Figure 6 displays the corresponding gross execution rates.

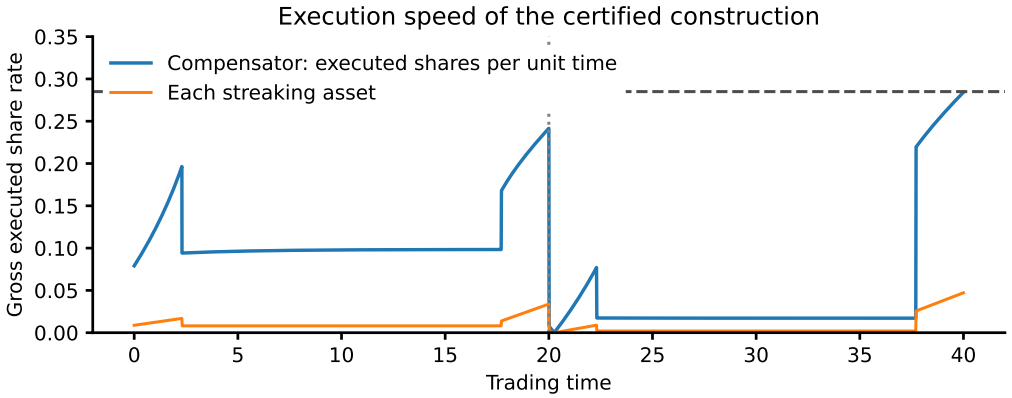


Figure 6. ***Gross execution rates in the certified path.*** *The largest computed gross share-rate is approximately* 0.28422*; rigorous interval rounding certifies a bound strictly below* 0.285*, without instantaneous pulses. The cap is per asset, before netting buys and sells.*

8. Discussion and model extensions.

8.1. Economic accounting and execution burden. Recovery has two roles: for the price-level dynamics in Assumption 2.1, it enters the curvature-based exclusion argument, but it also makes nontrivial terminal displacement possible under dollar closure. If $\rho = 0$, $\mathrm{d}\Phi(S_i) = \mathrm{d}X_i$ and $X_i(T) = 0$ implies $S_i(T) = F$ (Appendix A). For $\rho > 0$, the identity (2.11) shows that the nonnegative terminal potential and cumulative recovery dissipation are financed by a reduction of net share inventory, valued at the fixed benchmark. For any terminally dollar-closed path γ, define this benchmark-valued inventory consumption by

$$\mathcal{K}(\gamma) := F \sum_i \big[\theta_i(0) - \theta_i(T)\big] = \sum_i E(S_i(T)) + \rho \sum_i \int_0^T \frac{(S_i(t) - F)^2}{g(S_i(t))}\,\mathrm{d}t \geq 0.$$

This is a well-defined *model accounting cost*, not a measured implementation shortfall or the realized profit of a named market participant. It does not include spreads, fees, external funding, or a counterparty balance sheet, and it must not be conflated with a share-flat trading round trip. The inventory inequality of Section 5 is a necessary resource constraint, not a sharp optimization formula.

As a direct bound for the ten-asset certificate, $E(S) = (S-1)^2/2$ and the certified terminal compensator lower endpoint $y(40) > 1.4047874$ give

$$\mathcal{K}(\gamma) \;\geq\; \sum_i E(S_i(40)) > \frac{9}{2}(1 - 0.9409)^2 + \frac{1}{2}(1.4047874 - 1)^2 > 0.0976.$$

The bound uses only the certified terminal interval; the full recovery integral is not interval-certified here. Let $\Gamma_{\alpha,\delta}$ denote the admissible paths satisfying the terminal crisis, pairing, closure, and execution restrictions. A natural extension is to optimize

$$\mathcal{K}_{\min}(\alpha, \delta) = \inf_{\gamma \in \Gamma_{\alpha,\delta}} \mathcal{K}(\gamma).$$

Determining this infimum, or jointly minimizing cost and the gross rate, is a separate optimal-control problem; our necessary turnover bounds need not be sharp.

8.2. Dependence on resilience, impact, and the loss metric. Our global elasticity conditions rely jointly on the local response g and linear *price-level* restoration toward F. Under volume displacement recovery as described in Section 2, $\dot{\Phi}|_{\mathrm{vol}} = -\rho SV(S)$ replaces the specific drift used in the tangent comparison; the $e \geq 1$ and $e \geq 2$ theorems are therefore not asserted for that alternative model. A more general theory would classify which restoring fields preserve analogous barrier regions. Neither the order-book square-root metaorder law nor the exponent of cumulative price impact can be directly identified with our $e(S) = Sg'(S)/g(S)$: observed cumulative impact depends on order size, execution history, and replenishment, whereas e differentiates an assumed *local per-share* response with respect to the price state [1, 7, 13].

The severity of Market C is also a boundary result: logarithmic loss diverges as a positive price tends to zero, while constant g yields finite coordinate distance and finite inventory potential at that boundary. Fixed strictly positive price floors, realistic depth deterioration, rate caps, and trade halts can all limit its quantitative significance without invalidating the mathematical existence theorem. Accordingly, Section 6 establishes only a quadratic *small-rate* order, and not a large prescribed crash under arbitrarily tight constraints.

The definition $\mathcal{C}_\alpha = \min\{\overline{L}, Q_\alpha\}$ uses the equal-weight geometric price index as well as coverage. Metric choice matters: in the $p = 1$ market, paired-dollar trades give $\mathrm{d}\sum_i S_i|_{\mathrm{trade}} = (\eta/F)\sum_i \mathrm{d}X_i = 0$, and the common price-level recovery preserves the initial sum NF. Thus the arithmetic mean remains F even when a partial-coverage geometric-index crisis is reachable. Market-capitalization-weighted indices, arithmetically aggregated prices, rolling-peak losses, and different coverage criteria define different target sets. The present exclusion theorems should not be reinterpreted without checking the revised metric.

8.3. What the two-session certificate establishes. The construction in Proposition 7.6 is a constrained *existence witness* for one specified ten-asset market. Nine equal target prices lose 5.91% over two sessions, while the receiving asset has a high-precision reconstructed final price near $1.40516F$. The equal-weight geometric-index decline is approximately 2.06% (the mean log loss is approximately 0.02081), whereas the arithmetic mean price decreases by only about 1.27%. These different percentages measure different objects. The receiving asset must support unusually large temporary price excursions and a broad rolling band in this selected path. Table 2 varies the composition *holding the control fixed*; its non-$k = 9$ rows do not prove infeasibility of optimized alternative controls.

The bracket $0.044 < \bar{v}_* < 0.285$ combines a universal necessary inequality for the fixed model parameters with a certified feasible upper witness. Its width reflects the difference between an intentionally nonsharp barrier and one hand-designed construction, not a numerically established near-optimum. The interval certificate proves the stated pathwise constraints and ledger closure; it does not certify an optimal policy, an order queue, an empirically likely event, or a realized exchange-wide crash. The price-band protocol is deliberately *exchange-agnostic*; its 3% target bands and 25% compensator bands are chosen model parameters, not calibrated Tokyo Stock Exchange limits. For institutional contrast, TSE sets daily limits as tiered *absolute yen* amounts based on reference prices and conditionally widens limits after specified

sequences of limit-bound trading with little or no execution [18]. Empirical studies of actual price-limit regimes [9, 19] therefore do not validate the proportional-reset mechanics or the feasibility of this particular synthetic path. A genuine TSE application would require a separate venue-specific execution model.

8.4. Extensions and empirical tests. Several natural extensions are outside the scope of the present reachability classification. First, alternative resilience coordinates and nonlinear recovery operators could be analyzed for preservation or failure of the Φ-based barriers; this requires new comparison results, not a relabeling of ρ. Second, heterogeneous (g_i, ρ_i, F_i), varying network topology and nonseparable cross-impact replace the simple common-potential argument. For invertible matrix impact $\mathrm{d}\boldsymbol{S} = \mathsf{G}(\boldsymbol{S})\,\mathrm{d}\boldsymbol{\theta}$, an additive transaction potential would depend on exactness of the one-form

$$\omega = \mathbf{1}^{\top} \operatorname{diag}(\boldsymbol{S})\mathsf{G}(\boldsymbol{S})^{-1}\,\mathrm{d}\boldsymbol{S}$$

and on the compatible sign of its restoring drift. Matrix symmetry alone is not sufficient. Cross-impact no-dynamic-arbitrage restrictions [2, 25] and empirical multivariate impact [5] provide separate benchmarks.

Third, optimizing $\mathcal{K}_{\min}$, executed-dollar turnover, and the globally minimal constrained gross rate would connect structural reachability with optimal control and intermediary incentives, including strategic predation [6]. Proposition 5.2 aggregates across assets and does not use per-asset terminal dollar closure. When $X_i(T) = 0$, the constant-impact dynamics also give the subset-level identity

$$\frac{F^2 - S_i(T)^2}{2} = \rho \int_0^T S_i(t)(S_i(t) - F)\,\mathrm{d}t \qquad \text{for every } i.$$

Unlike the aggregate potential, this identity isolates each falling asset, but further work is needed to turn it into a sharper turnover lower bound. The squared-price dynamics, $\mathrm{d}(S_i^2) = -2\rho S_i(S_i - F)\,\mathrm{d}t + 2\eta\,\mathrm{d}X_i$, also have a trade term without the denominator $1/S_i$; their restoring drift remains nonlinear. Finally, an upper price bound $S_i \le U_{\max}$ together with the gross share-rate cap gives $\dot{M} \le NU_{\max}\bar{v}/2$. For fixed positive turnover, this prevents concentration at the single peak of the time kernel in (5.5) and permits a tighter weighted-turnover estimate. Combining these constraints into sharp subset-level and session-dependent bounds remains open. Fourth, empirical implementation would require identifying local impact g, price versus volume recovery, trade direction, accessible inventories, and counterparties from order-book and transaction data. A static book snapshot or daily OHLCV sequence does not identify these quantities. The market-microstructure-invariance approach [22] provides a scaling comparator, not a calibration already performed here. Extensions to stochastic flow or equilibrium selection could address probabilities of no-news crashes, which deterministic existence and exclusion results alone do not determine.

9. Conclusion. For a specified price-level resilience and paired-dollar execution protocol, the response-adapted coordinate identifies global barriers, feasible partial-coverage regimes, and a sharp full-coverage ceiling. Necessary dollar turnover and inventory are conveniently expressed in the generalized price coordinate and in units of local benchmark impact. For

constant impact on a bounded positive-price region, executed-dollar turnover also bounds the terminal quadratic price-potential deficit. A finite gross-rate cap changes the scale of attainable small systemic losses, and one independently interval-certified two-session witness satisfies the simultaneous rate, band, and dollar-ledger requirements. The economic accounting identity makes explicit the benchmark-valued inventory consumed by dollar-closed price pressure. These are structural and resource statements about an idealized market, not a probability, incentive mechanism, or empirical prediction of an exchange-wide crash. Alternative resilience laws, heterogeneous cross-impact and minimum-cost controls offer distinct mathematical directions for future work.

Code and numerical data availability. The research code, numerical data, certificates, and reproduction instructions are publicly available at https://github.com/Siyuan-chat/market-crash-reachability. The unchanged original computational baseline is associated with commit `ea831c4`; the frozen software archive is separately identified by DOI 10.5281/zenodo.23428183. Appendix G describes the interval certificate and identifies its implementation. Reproduction commands and software requirements are in `REPRODUCE.md`; the checksum manifest is `release/SHA256SUMS`. The underlying certificate was previously regenerated from source, with independent floating-point ODE and exact-rational consistency checks.

Declaration on AI-assisted tools. AI-assisted tools were used for language editing, manuscript organization, and assisting with verification and figure-generation scripts. The author takes full responsibility for all mathematical proofs, numerical certificates, and scientific content.

Appendix A. Settlement, coordinate changes, and variation bounds. The continuous response rule is interpreted along finite-variation execution paths. During an idealized pulse, physical time is frozen ($\mathrm{d}t = 0$) and the states move along the response ODE as the executing share size varies. The pulse's executed dollar transfer is the *integral* $\int S_i \,\mathrm{d}\theta_i$, not the beginning-of-pulse price times the final share increment. By continuity of $g > 0$, its endpoints can be joined by an admissible response path whenever the intervening price stays positive.

For one asset the differential rule reads

$$\mathrm{d}\theta = \frac{\mathrm{d}S + \rho(S - F)\,\mathrm{d}t}{g(S)}. \tag{A.1}$$

Decompose $S = F + (S - F)$ and integrate:

$$\int_0^T S \,\mathrm{d}\theta = F[\theta(T) - \theta(0)] + \int_F^{S(T)} \frac{u - F}{g(u)}\,\mathrm{d}u + \rho \int_0^T \frac{(S - F)^2}{g(S)}\,\mathrm{d}t,$$

where the initial price is F. This proves (2.11). For $S < F$ the primitive $E(S)$ in (2.10) is also nonnegative because the integration orientation reverses the sign of the negative integrand. Hence if $\theta(T) = \theta(0)$, the total executed cost cannot be negative.

Let $y = \Phi(S)$ and $z = \Psi(S)$, with

$$\Phi'(S) = \frac{S}{g(S)}, \qquad \Psi'(S) = \frac{1}{g(S)}.$$

Then

$$\mathrm{d}y = \mathrm{d}X - \rho\frac{S(S-F)}{g(S)}\,\mathrm{d}t, \qquad \mathrm{d}z = \mathrm{d}\theta - \rho\frac{S-F}{g(S)}\,\mathrm{d}t. \tag{A.2}$$

The recovery terms have signs opposite to $S - F$. Thus whenever $y < 0$ (equivalently $S < F$), the no-trade drift in y is positive; likewise for z. By separating the positive and negative variations in (A.2), a trajectory ending at $y(T) \le \Phi(s) < 0$ must accumulate at least $-\Phi(s)$ in negative executed-dollar variation. One way to see this is to start at the *last* time when y equals zero (or $t = 0$ if no later zero exists). On the remaining interval $y \le 0$ and the recovery drift is nonnegative. The negative transaction variation must therefore account for the entire fall to $y(T)$. The same argument in z gives the share-sale bound. Summing sales across qualifying assets counts each paired-dollar transfer at most once. If a terminally qualifying asset has inventory cap H_i and $X_i(T) = 0$, equation (2.11) implies

$$E(S_i(T)) \le F\left[\theta_i(0) - \theta_i(T)\right] \le FH_i.$$

Since $E(S)$ increases as S falls below F, $S_i(T) \le s$ implies $E(s) \le FH_i$. This proves Theorem 5.1.

When $\rho = 0$, $\mathrm{d}\Phi(S_i) = \mathrm{d}X_i$. Terminal dollar closure therefore implies $\Phi(S_i(T)) = \Phi(F) = 0$ and, by strict monotonicity, $S_i(T) = F$. Within the present separable response class, recovery makes the nontrivial dollar-closed endpoint effect possible.

Appendix B. Local two-pulse and three-pulse expansions. We document the expansion used in Propositions 4.3 and 4.2. Normalize local variations by $x = S/F - 1$ and $c = g(F)/F^2$. At $S = F$, the executed-dollar coordinate $y = \Phi(S)$ satisfies $\Phi(F) = 0$, $\Phi'(F) = F/g(F)$, and

$$\frac{\mathrm{d}S}{\mathrm{d}y} = \frac{g(S)}{S}, \qquad \frac{\mathrm{d}^2S}{\mathrm{d}y^2} = \frac{g(S)[Sg'(S) - g(S)]}{S^3}. \tag{B.1}$$

Hence, writing $e_0 = e(F)$, the price ratio after a small dollar-coordinate displacement y is

$$\frac{S}{F} = 1 + cy + \frac{1}{2}(e_0 - 1)c^2y^2 + O(y^3), \tag{B.2}$$

with constants uniformly bounded for paths in a compact neighborhood of F.

For the two-pulse experiment, the first impulse displaces y by hu_i at time zero. It produces the initial price response according to (B.2). Recovery over T multiplies the *level* displacement $S - F$ by $r = e^{-\rho T}$; the terminal pulse subtracts the same dollar amount hu_i from the executed-dollar coordinate. Expanding Φ and Φ^{-1} around F gives

$$L_i(T) = c(1-r)hu_i + \frac{c^2h^2}{2}\,\lambda(r, e_0)u_i^2 + O(h^3), \tag{B.3}$$

$$\frac{1}{N}\sum_i L_i(T) = \frac{c^2h^2}{2N}(1-r)(2-e_0-r)\sum_i u_i^2 + O(h^3),$$

where the individual second-order coefficient λ is not needed separately. The linear term vanishes in the mean because $\sum_i u_i = 0$. If at least k components are positive, those assets

have positive first-order losses and the kth-largest loss is positive for small h; the mean is positive under the stated time-curvature condition.

For the three-pulse experiment, the dollar-coordinate impulses at $0, T/2, T$ have amounts hu_i, $-(1+r)hu_i$, and rhu_i with $r = e^{-\rho T/2}$. Direct substitution and second-order expansion yield

$$L_i(T) = \frac{1-e_0}{2} c^2 h^2 u_i^2 r(1-r^2) + O(h^3), \tag{B.4}$$

with first-order terms canceling exactly. If $e_0 < 1$ and every $u_i \neq 0$, all terminal losses are positive for sufficiently small h. Since every pulse amount vector is zero-sum and the amounts cancel componentwise over the full horizon, the construction also satisfies the paired-dollar and terminal dollar-closure rules. The initial inventory requirement and turnover are finite for fixed h.

Appendix C. Uniform-resource severe drawdowns for partial coverage. We prove Theorem 4.1(C) constructively. Fix $N \geq 2$, set $m = N-1$, and choose the constant-response model $g(S) = \eta > 0$. We may assume the target crisis threshold δ is large, since a path exceeding a larger threshold automatically exceeds every smaller one. It is enough to produce m terminal prices $\varepsilon \downarrow 0$ with one remaining compensator asset, because every $k < N$ satisfies $k \leq m$.

C.1. A radial representation of paired-dollar trading. For $g \equiv \eta$,

$$\mathrm{d}(S_i^2) = -2\rho S_i(S_i - F)\,\mathrm{d}t + 2\eta\,\mathrm{d}X_i, \qquad \mathbf{1}^\top \mathrm{d}\boldsymbol{X} = 0. \tag{C.1}$$

Let $R = (\sum_i S_i^2)^{1/2}$. A paired-dollar trade keeps R fixed; a positive-price change at fixed R is implementable as a finite-variation path in the positive orthant of that sphere. In particular, interpolate the squared prices $Y_i = S_i^2$ linearly between two positive vectors of the same sum: every intermediate $Y_i > 0$ and $\sum_i \mathrm{d}Y_i = 0$, so $\mathrm{d}X_i = \mathrm{d}Y_i/(2\eta)$ is an admissible paired-dollar transfer. The corresponding share increment is $\mathrm{d}\theta_i = \mathrm{d}S_i/\eta$.

Fix $\beta = 1/5$. Choose $c > 0$ sufficiently small that, for every $b \in [0, \beta]$, the quantities

$$a = \sqrt{1 - b^2 - (m-1)c^2}, \qquad H = a + (m-1)c + b \tag{C.2}$$

satisfy $a > \max\{b, c\}$ and $bH < 1$. For example, $0 < c < 1/(10m)$ suffices. The constructed path uses $b \in (0, \beta)$; the endpoint $b = 0$ is used only by continuous extension in the root-bracketing argument below. During stage $j \in \{1, \dots, m\}$, fix the spherical direction

$$\boldsymbol{u}_j = (c, \dots, \underbrace{a}_{j}, \dots, c, b), \qquad \|\boldsymbol{u}_j\|_2 = 1. \tag{C.3}$$

Set $\boldsymbol{S}(t) = R(t)\boldsymbol{u}_j$ throughout the stage. To make it solve the price equations, use

$$R(t) = F\left[H + (\sqrt{N} - H)e^{-\rho t}\right], \qquad \dot{\theta}_i = \frac{\rho F}{\eta}\left(u_{j,i}H - 1\right). \tag{C.4}$$

Indeed, differentiating $S_i = u_{j,i}R(t)$ in the price equation gives (C.4) and

$$\dot{X}_i = \frac{\rho F R(t)}{\eta} u_{j,i}(u_{j,i}H - 1), \quad \sum_i \dot{X}_i = \frac{\rho F R(t)}{\eta}\left(H\sum_i u_{j,i}^2 - \sum_i u_{j,i}\right) = 0, \tag{C.5}$$

because $\sum_i u_{j,i}^2 = 1$ and $\sum_i u_{j,i} = H$.

C.2. Closing the compensator's dollar flow. At $t = 0$ begin with $F\mathbf{1}$ and apply a positive-price pulse at fixed radius $F\sqrt{N}$ to the first direction $\boldsymbol{u}_1$. Switch between stages $\boldsymbol{u}_j$ using similar fixed-radius pulses. At time T use a final fixed-radius pulse to reach

$$S_1(T) = \cdots = S_m(T) = \varepsilon, \quad S_N(T) = \sqrt{R(T)^2 - m\varepsilon^2}. \tag{C.6}$$

The terminal price vector is positive for all sufficiently small ε.

For constant g, integrating (C.1) gives the exact executed-dollar ledger

$$X_i(T) = \frac{S_i(T)^2 - F^2}{2\eta} + \frac{\rho}{\eta}\int_0^T S_i(t)[S_i(t) - F]\,\mathrm{d}t. \tag{C.7}$$

For the compensator, its stage weight is always b. Define

$$\mathcal{F}_\varepsilon(b) = \frac{R_b(T)^2 - m\varepsilon^2 - F^2}{2\eta} + \frac{\rho}{\eta}\int_0^T bR_b(t)[bR_b(t) - F]\,\mathrm{d}t, \tag{C.8}$$

where R_b depends continuously on b through (C.4). With c fixed, choose a sufficiently large finite T such that $\mathcal{F}_0(\beta) < 0$. Such a choice exists because $R_\beta(t) \to FH_\beta$ and $\beta H_\beta < 1$: the integral in (C.8) then has a strictly negative linear term in T, while the terminal term remains bounded. Here H_β denotes H evaluated at $b = \beta$. Fix this horizon. Since $R_0(T) > F$, we have $\mathcal{F}_\varepsilon(0) > 0$ for all sufficiently small $\varepsilon > 0$, whereas

$$\mathcal{F}_\varepsilon(\beta) = \mathcal{F}_0(\beta) - \frac{m\varepsilon^2}{2\eta} < 0.$$

Continuity therefore gives $b(\varepsilon) \in (0, \beta)$ for which

$$X_N(T) = \mathcal{F}_\varepsilon(b(\varepsilon)) = 0.$$

Both c and T are fixed independently of ε.

C.3. Closing every target asset's dollar flow. It remains to make the other m terminal dollar balances equal. Whenever an asset is in the high state a rather than the low state c, the difference in its recovery integrand is

$$W(t) = (a - c)R(t)\big[(a + c)R(t) - F\big]. \tag{C.9}$$

This is strictly positive: $(a+c)H = 1 + b(a-b) + mac + bc > 1$, and $R(t) \geq FH$. Consequently $W(t) > 0$ for $t \in [0, T]$. Choose a partition $0 = t_0 < \cdots < t_m = T$ such that

$$\int_{t_{j-1}}^{t_j} W(t)\,\mathrm{d}t = \frac{1}{m}\int_0^T W(t)\,\mathrm{d}t, \qquad j = 1, \ldots, m. \tag{C.10}$$

The continuous, strictly increasing cumulative integral makes this partition possible. Let target asset j occupy the high state in stage j and the low state otherwise. All target assets have the same initial price F, terminal price ε, and recovery integral in (C.7), hence identical terminal balances $X_i(T)$. Every pulse and every stage respects $\sum_i \mathrm{d}X_i = 0$; since $X_N(T) = 0$, the common target balance must also vanish. Therefore

$$X_i(T) = 0 \qquad \text{for every } i = 1, \ldots, N. \tag{C.11}$$

C.4. Uniform finite turnover and inventory. The path remains in the positive sphere of radius at most $F\sqrt{N}$, so $0 < S_i \le F\sqrt{N}$ throughout. There are $m+1 = N$ fixed-radius pulses, and each pulse can be implemented by monotone componentwise interpolation in squared-price coordinates. Consequently the cumulative paired-dollar pulse turnover has an explicit, though loose, upper bound depending only on N, F, η. For example, a safe bound is $N^3F^2/(4\eta)$. During recovery-controlled stages, $\sum_i |\dot{X}_i|/2 \le \rho N F^2/\eta$, giving

$$M(\gamma) \le M_0 := \frac{N^3F^2}{4\eta} + \frac{\rho N F^2 T}{\eta}. \tag{C.12}$$

Likewise, share sales over N pulses plus the continuous stages are bounded by

$$H_i^- \le H_0 := \frac{NF\sqrt{N} + \rho F T(\sqrt{N}+1)}{\eta}, \tag{C.13}$$

independently of ε. Starting with at least H_0 shares of each asset rules out short sales. The compensator's price stays bounded and positive; its log loss remains finite. The other m losses are $\log(F/\varepsilon) \to \infty$. Hence their kth order statistic and the average loss both diverge when $k \le m$. Thus $\mathcal{C}_\alpha(T) \to \infty$ with fixed finite M_0, H_0, proving Theorem 4.1(C).

Appendix D. Construction proving the sharp full-coverage bound. The necessary radial inequality was established in the proof of Theorem 4.4. For sufficiency, we construct an admissible family approaching the bound. Fix $0 < b < 1/\sqrt{N}$, and choose

$$a = \sqrt{1-(N-1)b^2} > b, \quad H = a + (N-1)b, \quad R_b(t) = F\left[H + (\sqrt{N} - H)e^{-\rho t}\right]. \tag{D.1}$$

Use the N positive unit-vector directions given by permutations of $(a, b, \dots, b)$. At time zero pulse from $F\mathbf{1}$ to $R_b(0)\boldsymbol{u}_1$, then follow $S_i(t) = u_{j,i}R_b(t)$ using the same paired controls as in (C.4). When changing directions, use finite fixed-radius positive squared-price pulses. At time T, make a final pulse to the equal-price point

$$S_i(T) = \frac{R_b(T)}{\sqrt{N}}, \qquad i = 1, \dots, N. \tag{D.2}$$

For the recovery integrals to be identical for every asset, define

$$W_b(t) = (a-b)R_b(t)\big[(a+b)R_b(t) - F\big]. \tag{D.3}$$

Here $(a+b)H = 1 + Nab > 1$, so $W_b(t) > 0$ for $t \in [0, T]$. Split the interval into N parts with equal integrals of W_b. In each part a different asset is the sole high-price coordinate. All N assets now have the same initial price, the same terminal price, and the same recovery integral in (C.7). Their final dollar balances are therefore identical; since all trades are paired, their sum is zero and each balance is exactly zero.

As $b \downarrow 0$, the terminal equal price in (D.2) tends to

$$\frac{F}{\sqrt{N}}\left[1 + (\sqrt{N}-1)e^{-\rho T}\right],$$

whereas $b \uparrow 1/\sqrt{N}$ yields $S_i(T) \to F$. Continuity in b implies that every strictly intermediate common terminal price is reachable by a positive-price, finite-turnover, fully dollar-closed path. The corresponding log-loss range is precisely $0 < \delta < \delta_{\mathrm{crit}}(N, T)$, completing the sharpness proof. The endpoint δ_{crit} itself would require the limiting direction with zero components and is excluded by strict price positivity.

Appendix E. Bounded-rate control: analytic details and audit protocol. For the three-bin control (6.14), the identities $\int f = 0$ and $\int e^{-\rho(T-t)} f(t)\,\mathrm{d}t = 0$ follow from equal bin widths and the geometric ratio $r = e^{-\rho T/3}$. Write $h_0 = g(F)/F$ and $h_1 = g(F)(e_0 - 1)/F^2$. The perturbation equations are $y' = -\rho y + h_0 f$, $z' = -\rho z + h_1 f y$. Since $y(0) = y(T) = 0$ and $(e^{-\rho(T-t)})' = \rho e^{-\rho(T-t)}$, one integration by parts gives the exact identity

$$\int_0^T e^{-\rho(T-t)} f(t) y(t)\,\mathrm{d}t = \frac{\rho}{2h_0} \int_0^T e^{-\rho(T-t)} y(t)^2\,\mathrm{d}t. \tag{E.1}$$

The resulting coefficient is $\rho/2$. Substituting $h_1/h_0 = (e_0 - 1)/F$ and $L_i = -\log(S_i/F)$ proves (6.15). At $p = 1$ the quadratic coefficient vanishes; this is consistent with the all-asset barrier but does not exclude a partial-coverage loss under a different bounded signal. The independent verifier integrates the nonlinear ODE directly with the dollar-rate signal, computes gross share execution before netting, and compares the finite-amplitude loss with (6.15).

A high-precision reconstruction of the severe pulse construction confirms its terminal-dollar accounting. Bounded-rate controls constitute a more restrictive admissible class, whose local construction and barrier are established separately.

Appendix F. Numerical experiments and reproducibility. This appendix documents deterministic price paths constructed under the exact marginal-price settlement protocol. For the severe-crash construction we use $g(S) = 1$, $F = \eta = \rho = 1$. The code selects a positive price direction and numerically solves for the compensator direction and time partitions; between switches the radius and ledger have analytic primitives. At every transaction time the paired executed-dollar sum is zero, and at maturity *each* signed-dollar ledger closes to numerical accuracy.

Table 3

Constructed paths with severe partial-coverage losses. The reported turnover is sufficient for these paths; the minimum turnover is unknown. All rows use target log loss $L = 5$ and parameters normalized to unity.

N	k	T	$\mathcal{C}_\alpha(T)$	Turnover M	Max. sale shares
3	2	3	3.3168	2.6179	3.9320
12	11	4	4.5781	38.6861	8.0804
20	19	5	4.7477	104.1750	10.1906

Table 4 reports the effect of increasing the target log loss for $N = 3$, $k = 2$, and $T = 3$.

Table 4

Effect of increasing target log loss for the fixed $N = 3$, $k = 2$, $T = 3$ construction.

Target log loss L	Systemic loss $\mathcal{C}_{2/3}(T)$	Turnover M	Maximum cumulative share sales
3	1.9845	2.6128	3.9365
5	3.3168	2.6179	3.9320
10	6.6501	2.6179	3.9319
20	13.3168	2.6179	3.9319

The plotted experiment uses a finite, high-resolution sampling grid only to display continuous recovery segments; the underlying segment values and cash-flow integrals are evaluated using analytic expressions, and terminal closure is checked separately. For the $N = 12$, $L = 5$ construction, the maximum terminal dollar-ledger residual under double precision is less than 10^{-13} in the distributed script. Numerical tolerance is evidence of correct accounting implementation, not a substitute for the constructive proof in Appendix C.

The numerical panels use model-generated inputs. Implementation details and reproduction commands are provided in the repository documentation.

Appendix G. A reproducible interval certificate for the two-day construction. This appendix specifies a computer-assisted proof in a form that can be independently audited. The code `verification/certify_streak_interval.py` uses Python integers, rational arithmetic, and a fixed-point interval representation with scale factor $Q = 10^{32}$, corresponding to resolution Q^{-1}; every multiplication and division is rounded outwards. The rational endpoint bracket for h in (7.18) has strictly opposite polynomial signs and the polynomial is strictly increasing on that bracket, so it contains the unique relevant exact root.

On each linear target segment, write $x(t) = a + vt$ in local segment time and $u(t) = x(t)[v + \rho(x(t) - 1)]$. The compensator equation is the scalar nonautonomous ODE $\dot{y} = f(t, y) := -\rho(y - 1) - 9u(t)/y$. Its derivatives away from the finite set of target slope-change times are

$$f_y = -\rho + \frac{9u}{y^2}, \qquad f_t = -\frac{9u'}{y}, \qquad y'' = f_t + f_y f, \qquad u' = v[v + \rho(2x - 1)].$$

The exact target plateaus and slopes yield conservative whole-segment enclosures $|u| < 0.107$ and $|u'| < 0.041$. On the bootstrap strip $0.70 \le y \le 1.60$, these imply $|f| < 1.70$ and

$$|y''| \le \frac{9|u'|}{0.70} + \left(\rho + \frac{9|u|}{0.70^2}\right)\left(0.60\rho + \frac{9|u|}{0.70}\right) < 5.$$

A tighter value of this last bound is recomputed with outward-rounded rational arithmetic on each of the six target segments. On a fixed grid of $h_0 = 1/2000$, a scalar comparison principle is combined with the one-step Taylor enclosure

$$y(t + h_0) \in y(t) + h_0 f(t, y(t)) + [-B_j h_0^2/2, B_j h_0^2/2],$$

where B_j is the certified segment remainder bound. If the beginning state belongs to $[l, u]$, comparison of scalar ODE solutions gives the next lower bound from $l + h_0 f(t, l) - B_j h_0^2/2$

and the upper bound from $u + h_0 f(t,u) + B_j h_0^2/2$, each rounded outwards. A local $|f| < 1.70$ excursion bound encloses the solution between grid points and verifies, inductively, that it stays in the bootstrap strip. The bound is reinitialized at each piecewise-linear slope discontinuity without taking an artificial step across it.

There are exactly $80,000$ certified grid steps. For the path in Proposition 7.6, the machine-checkable output gives $0.7534859 < y(t) < 1.223386$ over the first day and $1.0301664 < y(t) < 1.406405$ over the second. It also encloses $1.2219388 < y(20) < 1.2225356$ and $1.4047874 < y(40) < 1.4055543$, establishing the moving 25% second-session limits and a terminal interval containing the high-precision reference value 1.405160822. For the nine target assets, the target coordinates follow exact line segments, with their daily lows attained at the prescribed closes and their daily highs below the permitted ceilings. Within-step interval ratios enclose both $|u|/x$ and $9|u|/y$ and give the gross-rate bound in (7.19). The same enclosures, with outward-rounded time sums, bound individual gross share sales and executed-dollar turnover, as claimed.

Finally, the terminal mean loss is $-\log[(0.9409)^9 y(40)]/10$. The assertion that it exceeds 0.020 is checked *without* relying on a floating-point logarithm: finite rational Taylor series provide lower and upper bounds on $e^{0.2}$ and verify $e^{0.2}(0.9409)^9 \sup y(40) < 1$. A second rational check at $e^{0.21}$ confirms the mean loss remains below 0.021. These assertions certify feasibility of the stated piecewise-linear path and its compensator ODE.

References.

[1] A. Alfonsi, A. Fruth, and A. Schied, *Optimal execution strategies in limit order books with general shape functions*, Quant. Finance, 10 (2010), pp. 143–157, https://doi.org/10.1080/14697680802595700.

[2] A. Alfonsi, F. Klöck, and A. Schied, *Multivariate transient price impact and matrix-valued positive definite functions*, Math. Oper. Res., 41 (2016), pp. 914–934, https://doi.org/10.1287/moor.2015.0761.

[3] A. Alfonsi and A. Schied, *Optimal trade execution and absence of price manipulations in limit order book models*, SIAM J. Financial Math., 1 (2010), pp. 490–522, https://doi.org/10.1137/090762786.

[4] A. Alfonsi, A. Schied, and A. Slynko, *Order book resilience, price manipulation, and the positive portfolio problem*, SIAM J. Financial Math., 3 (2012), pp. 511–533, https://doi.org/10.1137/110822098.

[5] M. Benzaquen, I. Mastromatteo, Z. Eisler, and J.-P. Bouchaud, *Dissecting cross-impact on stock markets: An empirical analysis*, J. Stat. Mech. Theory Exp., 2017 (2017), p. 023406, https://doi.org/10.1088/1742-5468/aa53f7.

[6] M. K. Brunnermeier and L. H. Pedersen, *Predatory trading*, J. Finance, 60 (2005), pp. 1825–1863, https://doi.org/10.1111/j.1540-6261.2005.00781.x.

[7] F. Bucci, M. Benzaquen, F. Lillo, and J.-P. Bouchaud, *Crossover from linear to square-root market impact*, Phys. Rev. Lett., 122 (2019), p. 108302, https://doi.org/10.1103/PhysRevLett.122.108302.

[8] G. Cespa and T. Foucault, *Illiquidity contagion and liquidity crashes*, Rev. Financ. Stud., 27 (2014), pp. 1615–1660, https://doi.org/10.1093/rfs/hhu016.

[9] T. Chen, Z. Gao, J. He, W. Jiang, and W. Xiong, *Daily price limits and destructive market behavior*, J. Econometrics, 208 (2019), pp. 249–264, https://doi.org/10.1016/j.jeconom.2018.09.014.

[10] R. Cont and E. Schaanning, *Fire sales, indirect contagion and systemic stress testing*, Working Paper 2/2017, Norges Bank, 2017, https://www.norges-bank.no/aktuelt/publikasjoner/Working-Papers/2017/22017/.
[11] R. Cont and L. Wagalath, *Running for the exit: Distressed selling and endogenous correlation in financial markets*, Math. Finance, 23 (2013), pp. 718–741, https://doi.org/10.1111/j.1467-9965.2011.00510.x.
[12] F. Cordoni and F. Lillo, *Instabilities in multi-asset and multi-agent market impact games*, Ann. Oper. Res., 336 (2024), pp. 505–539, https://doi.org/10.1007/s10479-022-05066-8.
[13] G. Curato, J. Gatheral, and F. Lillo, *Optimal execution with non-linear transient market impact*, Quant. Finance, 17 (2017), pp. 41–54, https://doi.org/10.1080/14697688.2016.1181274.
[14] J. Gatheral, *No-dynamic-arbitrage and market impact*, Quant. Finance, 10 (2010), pp. 749–759, https://doi.org/10.1080/14697680903373692.
[15] G. Gennotte and H. Leland, *Market liquidity, hedging, and crashes*, Amer. Econ. Rev., 80 (1990), pp. 999–1021, https://ideas.repec.org/a/aea/aecrev/v80y1990i5p999-1021.html.
[16] J. Huang and J. Wang, *Liquidity and market crashes*, Rev. Financ. Stud., 22 (2009), pp. 2607–2643, https://doi.org/10.1093/rfs/hhn086.
[17] G. Huberman and W. Stanzl, *Price manipulation and quasi-arbitrage*, Econometrica, 72 (2004), pp. 1247–1275, https://doi.org/10.1111/j.1468-0262.2004.00531.x.
[18] Japan Exchange Group, *Daily price limits: Trading rules of domestic stocks*. https://www.jpx.co.jp/english/equities/trading/domestic/06.html. Accessed October 2026.
[19] K. A. Kim and S. G. Rhee, *Price limit performance: Evidence from the Tokyo Stock Exchange*, J. Finance, 52 (1997), pp. 885–901, https://doi.org/10.1111/j.1540-6261.1997.tb04827.x.
[20] A. Kirilenko, A. S. Kyle, M. Samadi, and T. Tuzun, *The flash crash: high-frequency trading in an electronic market*, J. Finance, 72 (2017), pp. 967–998, https://doi.org/10.1111/jofi.12498.
[21] A. S. Kyle, *Continuous auctions and insider trading*, Econometrica, 53 (1985), pp. 1315–1336, https://doi.org/10.2307/1913210.
[22] A. S. Kyle and A. A. Obizhaeva, *Large bets and stock market crashes*, Rev. Finance, 27 (2023), pp. 2163–2203, https://doi.org/10.1093/rof/rfad008.
[23] A. A. Obizhaeva and J. Wang, *Optimal trading strategy and supply/demand dynamics*, J. Financial Markets, 16 (2013), pp. 1–32, https://doi.org/10.1016/j.finmar.2012.09.001.
[24] D. Romer, *Rational asset-price movements without news*, Amer. Econ. Rev., 83 (1993), pp. 1112–1130, https://www.nber.org/papers/w4121.
[25] M. Schneider and F. Lillo, *Cross-impact and no-dynamic-arbitrage*, Quant. Finance, 19 (2019), pp. 137–154, https://doi.org/10.1080/14697688.2018.1467033.
[26] Y. Yang, *Finite horizon optimal execution with bounded rate of transaction*, Stoch. Models, 35 (2019), pp. 469–495, https://doi.org/10.1080/15326349.2019.1621760.